The Spatial Cost of Formality: Street Vendors' Preferences in India

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Abstract

A state cannot grant legal status to a firm trading on public land without also deciding where it may trade. For these firms formality is offered at a location rather than at a price, and across the developing world it is delivered through designated vending zones that are commonly empty. We price what relocation costs such a firm, using a stated preferences experiment embedded in an original geo-located survey of 2,922 street vendors across 206 marketplaces in Patna, India, where the Street Vendors Act of 2014 makes certification free and conditions the legal right on location instead. Formality is valued: offered a zone at an unchanged commute, the average vendor would accept a rent of up to 4.5 per cent of monthly sales, and 69 per cent accept a free spot on those terms. What a zone is worth varies widely across vendors, and the 27 characteristics we observe shift where a vendor sits without explaining why vendors differ, so the ones a zone would reach cannot be identified in advance. Distance consumes that value and does so unevenly. The acceptable rent falls to 2.1 per cent of sales after a one kilometre longer commute and to 0.9 per cent after three, more than half the value going in the first kilometre alone, and reaches zero at about 3.8 kilometres, beyond which no rent is acceptable at all. Information raises the value of a zone, lifting take-up by about 3 percentage points from a baseline at which 1.3 per cent of vendors can name the Act, but changes none of these trade-offs. Zones sit empty not because the state has failed to offer formality or because vendors do not want it, but because conferring it requires moving the firm, and moving it costs most of what the legal status is worth.

JEL Classification: O17, R38, C25, O18, C93
Keywords: Informality, Firm Location, Relocation costs, Vending Zones, Stated Preferences

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1 Introduction

Around 2 billion people, or 60% of the global workforce, work in the informal sector. Governments in developing countries commonly attribute low fiscal capacity and low aggregate productivity to persistent informality, and have introduced policies that grant informal firms legal rights and social protection upon registration. Registration rates nonetheless remain low, and how much a given policy achieves depends on how it is designed and on how much the firms it targets want what it offers.

One sub-group of informal firms sits awkwardly in this framework. Street vendors are own-account firms that sell goods and services from carts, stalls, and ground displays in public spaces to which they have no legal right. Street vending accounts for 1.5 to 6.8% of total employment in low- and middle-income countries (Roever and Skinner 2016). Vendors are the final link in urban supply chains for inexpensive goods and produce, serving households along commuter corridors. For these firms a government cannot confer legal status without also deciding where they may lawfully trade. Formality is therefore offered at a specified location, and accepting it means relocating the firm.

Where vendors trade is contested. They concentrate on main roads and pedestrian walkways claimed simultaneously by vehicles and pedestrians, especially in dense cities, and over the past decade governments have debated intensively how to regulate the sector, in order to ease congestion, keep public spaces usable, and bring vendors onto a register and into the tax base. The debate has produced forceful policy, including Accra's decongestion by-laws, Bangkok's Return the Footpath to Pedestrians, Ho Chi Minh City's Civilized Streets, and Johannesburg's Operation Clean Sweep. Appendix A1 documents 54 such policies across 27 countries and six regions. Just over 70% of them establish vending zones, making the zone the most common instrument for formalising vendors. A vending zone is a demarcated, legally protected area in which certified vendors receive fixed-size spots for a monthly rent. Relocation to that zone is the condition on which legal status is granted, and it is a condition the standard instruments of formalisation policy do not address.

Those instruments, and the costs they target, have been studied closely. The administrative and monetary costs of becoming formal and of remaining formal are well documented, and the consensus is that removing the fixed costs of registration raises formality by less than reducing the recurring costs of remaining formal or raising the cost of informality through enforcement (Bruhn and McKenzie 2014; Rocha, Ulyssea, and Rachter 2018; Farias and Hsu Rocha 2026; Andrade, Bruhn, and McKenzie 2016; Ulyssea 2020; Campos, Goldstein, and McKenzie 2023), though which margin binds remains contested. Far less attention has been paid to the relocation costs that formality can carry, or to firms operating on public land. Where land appears at all, it appears as an eligibility condition rather than as a price. de Mel, McKenzie, and Woodruff (2013) offered informal firms in Sri Lanka payments worth up to two months of median profits to register, and firms operating on public land were 27 percentage points less likely to register, some of them abandoning the process after beginning it, because they could not produce the paperwork establishing a legal claim to the premises they occupied. Firms screened out on documentation never face a choice, so what formality is worth to them, and what they would give up for it, is never observed.

This paper measures the relocation costs that formalisation imposes on firms without premises, and the preferences of the firms the policy is aimed at. Focusing on India, we separately identify and value four potential barriers to relocation: location and commuting costs, rent, coordination failures among vendors and customers, and limited information about the policy. We thereby

identify the binding constraint and show how policy could be redesigned. In doing so we create new and detailed data on a group of informal firms about which little is known.

India's Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act, 2014 provides a useful setting. It requires every urban local body to survey and certify vendors and establish vending zones. Certification is free, issued upon survey, and does not require proof of a claim to the vendor's current site. The Act thus eliminates registration costs, the first standard explanation for informality, and makes relocation the only condition attached to legal status. Low take-up therefore points to the terms of relocation rather than to registration costs. We study the Act's implementation in Patna, the capital of Bihar, where approximately 30,000 vendors operate in public spaces. The Municipal Corporation has built several vending zones, all but one of which have zero occupancy.

That near-total non-relocation is also why observed behaviour cannot reveal how vendors value the terms on offer. Vendors rarely change marketplaces; the zones are too few and too empty for their attributes to vary meaningfully in the field; and the share of peers who relocate is jointly determined with each vendor's own decision. We therefore elicit preferences through a stated-preference experiment, which recovers preferences over attributes that do not vary in the field (McFadden 1974). We conducted an original geolocated survey of 2,922 vendors across 206 marketplaces in Patna, recording each vendor's location, rent, commute, and trading conditions. In the experiment each vendor chooses five times between remaining at the current site and moving to a hypothetical vending zone, whose distance from the vendor's residence, rent, and expected trading conditions are all defined relative to that verified status quo. Between choices, half the vendors in every marketplace receive a bundled treatment giving information about the Act and about the credibility of its enforcement. Section 3 describes the design and our measures to limit hypothetical bias.

We impose no functional form at the outset. Every attribute enters as a full set of level dummies, and the simpler forms we adopt are restrictions of that model that the data do not reject. Both commuting distance and rent are non-linear, so neither can be summarised by a single price. We then allow vendors to differ in how they value a zone and in what each additional kilometre of commute costs them. Our main estimates come from a mixed logit with these two random coefficients, reported alongside a fixed-effects conditional logit that imposes no assumption on how vendors differ; the two agree on every trade-off we report.[1] Two independent variance estimators, the clustered delta method and a 200-replication marketplace pairs bootstrap, agree to within a few hundredths across every quantity we report.

Vendors value formality. A municipality designating an existing site as a zone, so that the commute does not change, could charge the average vendor up to 4.5% of monthly sales, approximately INR xx per month at median sales, and 69% of vendors accept such a zone when the spot is free. Vendors value zones mainly for protection rather than productivity: among those choosing a zone, 86% cite protection from eviction or harassment and 60% cite reduced stress and uncertainty, whereas fewer cite expected footfall or amenities. This accords with evidence that registration generally does not raise productivity or earnings, and that its benefits arise primarily from reduced risk and access to social insurance (de Mel, McKenzie, and Woodruff 2013; Benhassine et al. 2018; Farias and Hsu Rocha 2026; Ulyssea 2020).

[1] Appendix **??** reports the alternative specifications. A richer specification, which also allows the cost of changing site to vary across vendors, fits better and is rejected against by the data at conventional levels. It leaves every quantity reported in this paper unchanged except the dispersion of the distance response over short moves, which the specification used here understates by a factor of about 2.5. Statements below about how widely vendors differ are therefore lower bounds.

That average conceals most of what is going on, and what it conceals is not something a municipality could anticipate. Once the 27 vendor characteristics we observe are already in the model, allowing vendors to differ is worth a further 760 held-out log points, against about 29 from the characteristics themselves. Those characteristics, which include earnings, education, tenure, type of pitch, and exposure to eviction, generate 15.5% of the total spread in the value of a zone, yet adding them reduces the unexplained variation by only 6.1%. They shift where a vendor sits on average without explaining why that vendor differs from others like them. A zone programme therefore cannot be directed at the vendors most likely to take it up, because they cannot be identified in advance from a survey. What a zone can be designed around is its terms.

Vendors can obtain this value, however, only by relocating, and relocation is costly because location is a principal input into their businesses: customers may not follow them, and vendors hold no title to their current sites that they can sell or use as collateral. The maximum rent the average vendor would accept falls from 4.5% of monthly sales at the current commute to 2.1% when the commute is one kilometre longer and 0.9% when it is three kilometres longer, and reaches zero at about 3.8 kilometres of additional commute. At the five-kilometre move the experiment offered, there is no rent, not even zero, at which the average vendor accepts. The cost is front-loaded: more than half the value of the zone is lost to the first kilometre alone, and the two kilometres after it together cost less than that first one. This is not status quo bias, because we find no comparable penalty for changes from current rent or marketplace size (Appendix A7). It is what the vendor gives up by leaving the site: only 14.9% of regular customers are expected to follow, and 79% of vendors expect lower customer volume at a new site.

The remaining three barriers account for much less. Rent bites from the first percentage point and most steeply between 1% and 2% of sales, a percentage point being worth about 1.0 km of commute below 1%, 1.8 km between 1% and 2%, and 0.6 km above 2%; but at the current commute, acceptance falls only from 69% when the spot is free to 61% at a rent of 2% of sales. We find no detectable effect of the expected share of relocating peers, the explanation most commonly advanced for unoccupied formal space (Cooper and John 1988; Brock and Durlauf 2001): the estimate is zero to three decimal places, and the confidence interval rules out effects larger than about 0.08 km of commuting distance per 10 percentage points over the range the randomised shock delivered. Information raises the relative value of the zone without relaxing the relocation constraint, lifting take-up by about 3 percentage points at every zone design from a baseline at which only 1.3% of vendors can name the Act.

Two further margins we bound rather than price. The number of spots a zone offers has no detectable effect on relocation, so building larger zones is not a remedy. Spots in Bihar's zones measure either 6 by 4 or 5 by 3 feet, less space than most vendors currently occupy, and because our choice tasks hold scale of operation fixed, stated willingness to move assumes an accommodation the zones as built cannot provide.

These results contribute to the formalisation and relocation literatures. To our knowledge, this is the first paper to value the barriers to formalisation facing firms without premises on a common scale, each vendor's monthly sales, allowing the barriers to be ranked rather than merely listed. The ranking is not the one the policy debate assumes: distance first, rent second, information third, and coordination indistinguishable from zero. The paper also values land, which the formalisation literature generally treats only as an eligibility condition. Because the Act removes the documentary requirements that would otherwise exclude firms without land, we can directly estimate the cost of exchanging location for legal status, and we observe the preferences of firms that such requirements would ordinarily screen out before any choice is

made.

The relocation literature has primarily studied households. Evaluations of slum relocation find that destination location matters as much as housing quality and that a better dwelling in a worse location may leave households no better off (Takeuchi, Cropper, and Bento 2008; Barnhardt, Field, and Pande 2017; Harari and Wong 2026). Evidence is mixed, however: in Addis Ababa, subsidised relocation to peripheral housing had high take-up and no adverse labour-market effects (Franklin 2025; Agness and Getahun 2024). Studies using discrete-choice methods to identify what relocation sites must offer have relied on moves that occurred (Takeuchi, Cropper, and Bento 2008; Lall, Lundberg, and Shalizi 2008). For households, location is partly a consumption choice, and movers can change employers to offset some losses. For firms, location is an input into production, and no comparable substitution is available. By experimentally assigning destination attributes rather than observing them, we separate the cost of moving at all from the cost of distance. These components imply different policies: the first can only be avoided, whereas the second can be compensated.

The results also delimit what policy design can achieve. Where an existing vending site can be designated as a zone, vendors receive legal status without incurring the cost of moving, and our estimates suggest that this is close to the only viable option. Designation is not always feasible, because the congestion and land scarcity motivating regulation also constrain zone placement, and nine in ten sampled vendors trade on main roads. Where relocation is unavoidable, little of the distance can be bought back through rent, since a zone one kilometre away can charge at most 2.1% of sales to begin with. Zones remain empty not because the state fails to offer formality or because vendors do not value it, but because granting legal status to firms without premises requires relocation. Relocation consumes more than half the value of legal status within the first kilometre and all of it by about four.

The remainder of the paper is organised as follows. Section 2 describes the institutional setting, sampling frame, and survey. Section 3 presents the stated-preference design, discrete-choice model, and information experiment. Section 4 reports the results, and Section 5 discusses their implications and limitations.

2 Data

2.1 Institutional Context

In India, street vending is governed by the Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act, 2014 (SVA henceforth). The Act requires every urban local body to survey the street vendors in its jurisdiction, issue certificates of vending, and designate vending zones to which vendors are to be allocated; state governments frame the rules and schemes under which those obligations are discharged. Vendors accommodated in these zones receive legal recognition and protection from eviction, rights that are absent in their current informal locations, and recognition under the SVA also makes them eligible for complementary government schemes, most notably the Pradhan Mantri Street Vendor's AtmaNirbhar Nidhi (PM SVANidhi) scheme, which provides subsidised collateral-free loans. Implementation is overseen by Town Vending Committees (TVCs) at the municipal level, with state governments retaining flexibility over licensing terms, fee structures, and renewal conditions. This decentralised approach has produced substantial variation in implementation across cities.

The Act's core mechanism is spatial, and it is worth being precise about which costs it removes and which it relocates. Certification carries no fee, and the vendor files nothing: the munici-

pality conducts the survey at the marketplace and registers vendors there. Nor does eligibility rest on documenting a claim to the site occupied, a requirement that elsewhere excludes firms trading on public land. The fixed cost of entering the formal sector, on which a substantial part of the formalisation literature has focused, therefore falls on the state rather than on the firm. What remain are recurring obligations: the scheme framed under the Act provides for a monthly charge for a spot inside a vending zone, and a certificate of vending must be reactivated at five-yearly intervals against a small fee (India 2014). Legal status is thus conditioned on neither payment at entry nor documented tenure, but on where the vendor trades.

Where those zones are sited is not a residual administrative decision. The stated motivation of policymakers in relocating vendors is to relieve the congestion of carriageways and pedestrian walkways, so the sites on which vendors concentrate are precisely the sites a municipality is seeking to clear. The same density constrains the alternatives, since a zone requires a plot of a given size and few such plots are available in the built-up centre of a congested city. Both considerations push the location of a zone away from the corridors on which vendors currently trade, and a municipality implementing the Act is correspondingly constrained in the offer it can make.

We selected Patna, the capital city of Bihar, as the study site for three reasons. First, its entire urban area, spanning 75 wards, is governed by a single statutory authority, the Patna Municipal Corporation (PMC). All street vendor regulation, including licensing, enforcement, and policy implementation, is handled through one set of rules. This unified jurisdiction eliminates the confounding effects of inter-agency variation present in cities like Delhi, where multiple municipal corporations operate with potentially different enforcement practices, and simplifies both data collection and the interpretation of vendor behaviour.

Second, Patna's street vending landscape is shaped by commuter-led demand rather than by tourism. Vendors serve a stable local customer base of office workers, daily-wage earners, and households, selling necessities, predominantly produce and food for immediate consumption, priced for the city's purchasing power, and their stalls are long-established and embedded in kin-based social networks. This low-churn setting is well suited to studying the expectations and social ties that shape relocation decisions, without the confounding variation a transient vendor population would introduce.

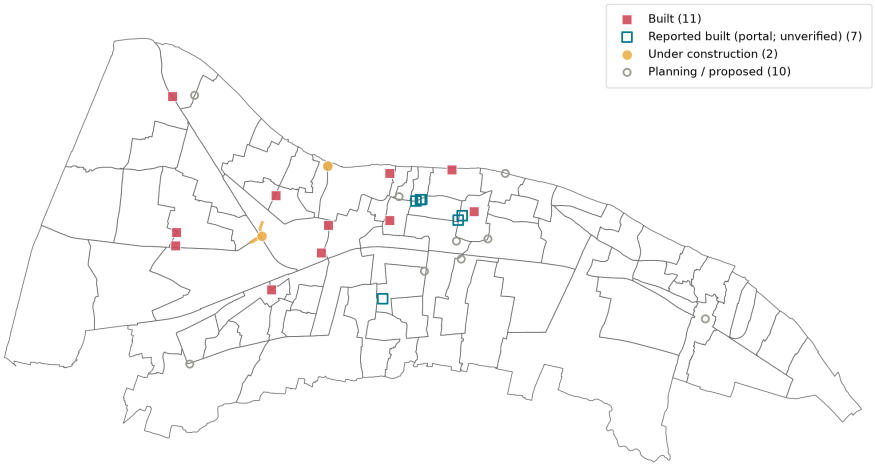
Third, the SVA's implementation failures are particularly visible in Patna. Bihar is among India's most populous and least developed states, and the street vending sector in Patna is large, at approximately 30,000 vendors according to PMC records. Under the SVA, PMC has constructed multiple vending zones, yet implementation has stalled: all but one record zero occupancy, and the exception is only a quarter utilised. Figure 1 shows their locations. The zone designs also fix the space allocated to each vendor spot. Under the Bihar implementation of the Act, each spot measures either 6 by 4 feet or 5 by 3 feet, a dimension we return to in Section 4.5.

2.2 Primary Survey

Data collection proceeded in three stages: marketplace listing, vendor recruitment, and the main survey. Patna is administratively divided into 75 wards. We selected 40 wards as the focus of the project based on vendor concentration, using the 2021 National Urban Livelihoods Mission (NULM) survey of vendors obtained from PMC. We geocoded potential vending locations, constructed vending zones, and overlaid these with ward boundaries to identify the 40 wards with the highest concentration of street vendor activity.

Marketplace Listing. We conducted a comprehensive listing exercise of marketplaces in these

Figure 1: Proposed and Constructed Vending Zones across the City



Note: Officially recorded street vending zones in Patna, by construction stage. Thirty-three appear in a government record: a SULM portal entry, the 2021 Urban Development & Housing Department sanction (Memo 11913/2021), or a record of construction. Thirty are plotted, of which 11 are built, 2 are under construction and 10 are in planning; a further 7 are reported built on portal entries whose status field and status text disagree. The remaining three could not be located. Coordinates come from the SULM portal (10) and geocoded landmarks (20). One zone is designated as a stretch of road rather than a point and is drawn along the road.

40 wards in August 2024. The 40 study wards contain approximately 17,000 static vendors, the substantial majority of Patna's stationary vendor population. During the listing exercise, we recorded the geolocation of each marketplace, characteristics of the surrounding area, and an approximate count of vendors operating in each market. We define a marketplace as any public space or location where more than 10 vendors are clustered selling goods or services. We define a street vendor as a person, typically self-employed, selling goods or services and operating their business from a temporary built-up structure at a fixed location in public spaces such as streets, lanes, sidewalks, footpaths, pavements, public parks, or any other public place. Importantly, we exclude hawkers and mobile vendors from this definition, focusing instead on vendors with fixed locations. This is narrower than the ILO definition given in Section 1, which admits mobile vendors. The restriction follows from the experiment: the choice tasks price attributes of a fixed location, and rent, number of spots and commuting distance have no counterpart for a vendor who does not occupy one. Our project specifically targets produce and street food vendors, who together comprise 66% of the street vending sector according to EBRD informal enterprise surveys. The listing exercise identified 375 marketplaces across the 40 wards, which were reduced to 206 analytical marketplaces in two steps. First, 43 marketplaces were excluded prior to the survey: their primary product was not produce or street food, and one market had been declared a commercial zone on a first-come-first-served basis, leaving 332 shortlisted marketplaces. Second, 126 of these were merged into nearby markets to protect the information experiment. Because treatment is assigned within marketplaces, contamination is governed by the time between a treated vendor being surveyed and a control vendor nearby being surveyed. Merging adjacent clusters into a single marketplace meant each was surveyed within a single day, leaving hours rather than days for information to travel; had they been retained as separate units, they would have been visited on different days, and vendors in the later cluster could have heard about the Act from those surveyed earlier.

Recruitment and Survey Administration We returned to the 206 identified marketplaces between June and October 2025 to conduct surveys. The survey process proceeded in two stages. First, we conducted recruitment visits during peak marketplace hours to identify eligible vendors and schedule survey appointments at times when vendors were relatively free to

participate. We employed a systematic random walk recruitment strategy, approaching every third vendor in each marketplace who fit our project focus (produce or street food vendors operating from fixed locations). We recruited 3,499 vendors.

Second, we administered the main survey to recruited vendors within a few days of the recruitment. Surveys were conducted in-person and typically lasted 45 minutes. Out of those recruited vendors who agreed to participate, 2,935 participated in the main survey yielding an initial response rate of 84%. We exclude 13 observations with incomplete responses, leaving a final analytical sample of 2,922 vendors across 206 marketplaces. Survey respondents are broadly similar to survey non-respondents (Appendix A2).

Figure 2, panel (a) maps the geo-locations of all 2,922 vendors in the sample across Patna, revealing clusters distributed across the city. The spatial distribution of vendors reveals strong clustering along transportation corridors. Panel (b) maps vendors' reported residences, geocoded from the landmarks they named to the centroid of the surrounding locality. Residences retain a dense core in the central wards but extend substantially further east, indicating that a sizeable share of vendors commute inward to their stalls. Panel (c) shows the markets within the city that vendors name as their source of supply. Including 14 markets that are outside city boundaries, wholesale (produce) markets sit largely on the periphery of the built-up area, while ingredient markets (retail markets) are concentrated in the centre alongside the densest vendor clusters. This implies that bulk produce enters the city at its edges and is being redistributed inward. 89% of vendors live within 3 km of their place of business (Figure 3 (a)) and 91% of them report that they don't pay any fees for occupying their current business spot (Figure 3 (b)).

Table 1 presents descriptive statistics for the analytical sample. Vendors are on average 39 years old with 6 years of education; 88% are male and 96% belong to historically disadvantaged caste groups (87% Other Backward Castes, 10% Scheduled Caste/Tribe). For most individuals, street vending is not transitional work: 53% have been vending for over 10 years, and 95% work 7 days per week averaging 9 hours per day. Average daily profit is INR 619 with average daily sales of INR 2,991. 85% of our sample operates from stalls or carts and 90% along main roads or streets, leaving them highly exposed to eviction.

Figure 4 describes the licensing and enforcement environment in which these vendors operate. Panel (a) shows that 20% hold a certificate of vending, so certification under the SVA has reached a fifth of the sample. Panel (b) reports how often eviction drives (*halla gadis*) occur in the vendor's marketplace: 10% report that they never do, while 70% report at least one a month and 23% at least one a week. Panels (c) and (d) record personal exposure over the six months before the survey, with 41% reporting harassment from officials or the police and 33% reporting an eviction or confiscation of goods. Enforcement is routine rather than exceptional, and it reaches vendors directly rather than through fines or paperwork.

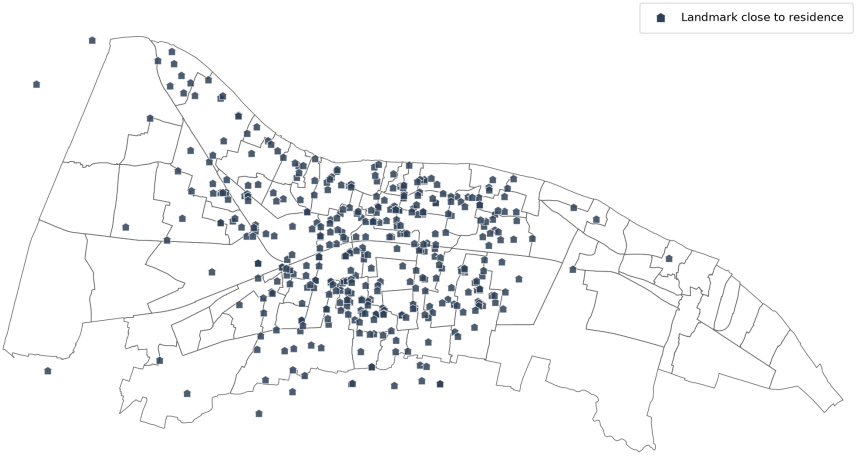
Set against this, awareness of the policy that would protect them is close to zero: only 1.3% could identify the Street Vendors Act (Figure 5). Only 7.7% of vendors were aware of any policy concerning street vending, and 6.3% knew the name of any policy. This combination, active enforcement pressure alongside near-zero policy awareness, motivates both the study setting and the information experiment described in Section 3.3.

Figure 2: Geo-location of vendors, their homes and sourcing markets

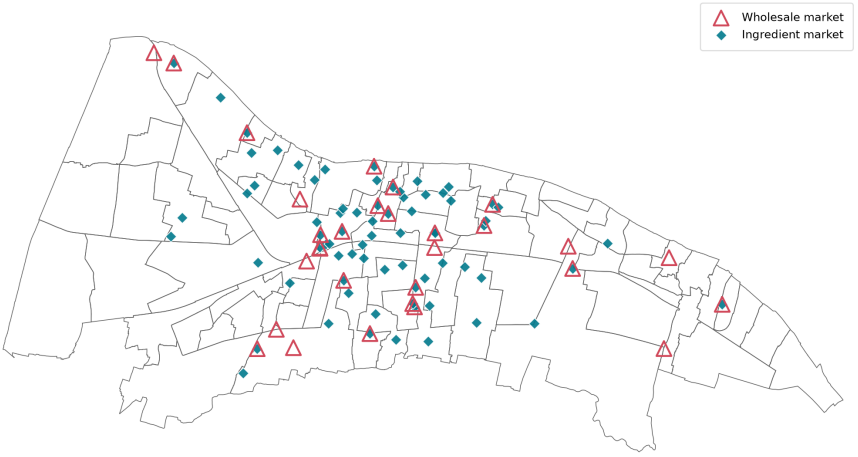
(a) Vendors' Business



(b) Vendors' Residence



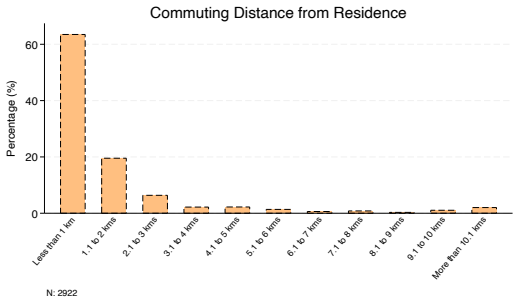
(c) Wholesale and Inventory Markets



Note: (a) Surveyed vendors across the 40 study wards. (b) Each marker is a distinct landmark reported as near a vendor's home, geocoded to its locality centroid; 412 of the 719 landmarks reported were geocoded confidently and fall within the mapped area. Residence locations are available for 2,374 of 2,922 vendors (81.2 per cent), plus 118 who live near their stall and take its coordinates; the remainder gave no answer or a landmark that could not be tied to a place name. (c) Markers are distinct markets named by vendors: outlined triangles are wholesale (produce) markets, filled diamonds ingredient markets. Fourteen markets outside the mapped area are included in all counts but not drawn. Market locations are available for 1,805 vendors (61.8 per cent), covering 84.0 per cent of fruit and 79.5 per cent of vegetable vendors but only 32.9 per cent of cooked-food vendors, who mostly source from neighbouring shops or take deliveries.

Figure 3: Status-quo Environment

(a) Reported Commuting Distance between Residence and Marketplace



(b) Share of Vendors Paying Informal Fees at Current Business Location

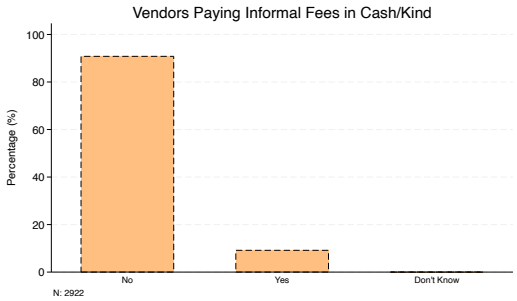
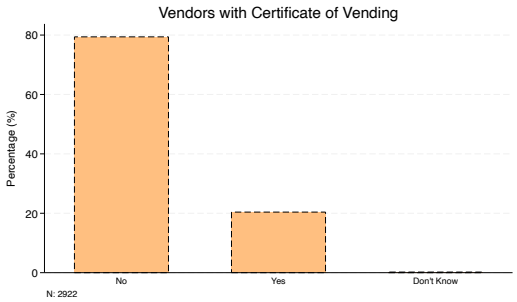
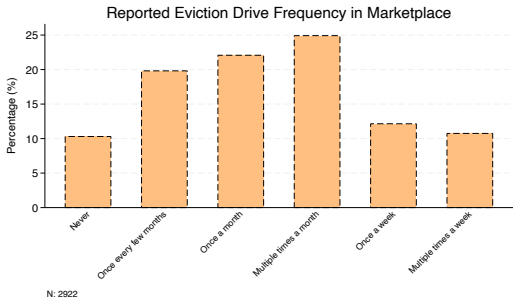


Figure 4: Licensing and Enforcement Environment

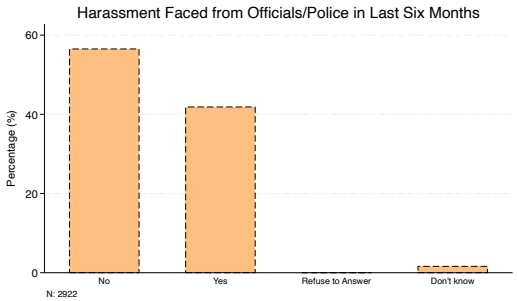
(a) Certificate of Vending



(b) Eviction Drives Frequency in Marketplace



(c) Harassment from Officials or Police



(d) Eviction or Confiscation of Goods

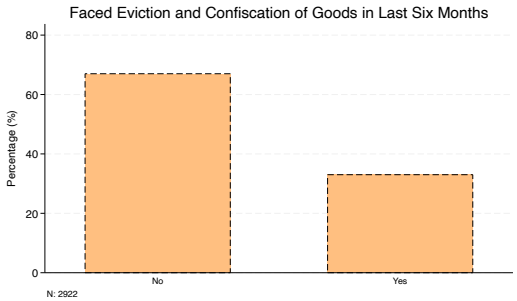


Figure 5: Awareness of Policies Among Street Vendors

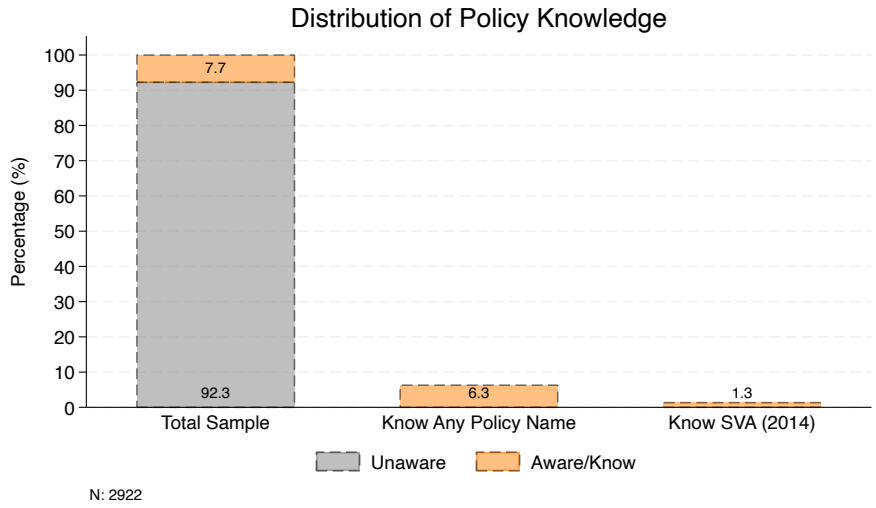


Table 1: Summary Statistics

Variable	Mean	SD
Age	38.64	12.76
Education	6.08	4.76
Male	0.88	0.32
Marital Status		
Married	0.82	0.38
Single	0.16	0.36
Widowed/Divorced	0.02	0.15
Caste Group		
Backward Caste	0.86	0.34
Scheduled Caste/Tribe	0.10	0.30
General	0.04	0.19
Product Category		
Fresh Produce	0.59	0.49
Cooked Food/Drinks	0.41	0.49
Premises Type		
On Main Roads or streets	0.90	0.30
On Pedestrian Walkways	0.10	0.30
Infrastructure Type		
Have Stalls/Carts	0.85	0.36
On the Floor	0.15	0.36
Other Business related		
Work 7 days a week	0.95	0.22
Hours of Operation	9.22	3.67
Have employees	0.32	0.46
Yesterday's Profits (INR)	618.53	626.31
Weekly Profits (INR)	4,259.65	4,056.31
Monthly Profits (INR)	18,458.50	17,577.33
Own Other Businesses	0.15	0.35
Never Changed Marketplaces	0.99	0.08
Years in Business		
Upto 1 year	0.06	0.24
1 to 5 years	0.21	0.41
5 to 10 years	0.20	0.40
More than 10 years	0.53	0.50

Note: N = 2922. All monetary units are expressed in Indian Rupees (INR). Monthly profits are constructed from reported weekly profits as weekly profits $\times$ 52/12. Profits were elicited for the previous day and the previous week only, since most vendors do not keep written accounts and longer recall windows are unreliable.

3 Empirical Strategy

Pricing the relocation the Act requires means recovering how vendors value the terms on which formality is offered: the rent charged, the distance from their residence, the number of spots, the competitive and demand conditions they expect to find, and the share of peers who also move. Three features of the setting prevent us from recovering these valuations from observed behaviour. First, 99% of vendors have never changed marketplace, so revealed preference data contain almost no variation in location choice. Second, the terms do not vary in the field: the zones PMC has constructed are few and almost entirely unoccupied (Section 2.1), so there is neither a spread of offers to compare nor take-up against which to price them. Third, the peer share is simultaneously determined, since the individual relocation decision depends on the share s of peers relocating while s is itself the aggregate of those decisions. Observational estimates would in any case confound the causal effect of peer behaviour with common unobservables driving both.

We address all three by assigning the terms of the offer rather than observing them. A stated preferences experiment presents each vendor with a sequence of hypothetical zones whose attributes are drawn from a designed distribution, so that rent, distance and the peer share vary independently of anything about the vendor. The approach is well established in transportation and environmental economics and has been applied to labour market choice by Mas and Pallais (2017). Two features of our implementation are intended to limit hypothetical bias: the scenarios are personalised, with every attribute expressed relative to each vendor's verified current conditions, which anchors the choices to circumstances the respondent already knows (Taylor and Cummings 1999); and vendors were told at the outset that their responses would inform PMC policy. Appendix A6.2 sets out these and the other design responses to hypothetical bias, inattention and fatigue. A randomised information treatment, described in Section 3.3, is embedded between choice rounds.

3.1 Stated Preferences Experiment

Cognitive Testing: The vignette was tested with approximately 50 street vendors in Patna before it was finalised. Given that many street vendors have low levels of literacy, the format had to be accessible to respondents regardless of their reading ability. We presented draft versions of the choice tasks and solicited feedback on the framing of the question, the language used to describe attribute levels, and the visual presentation of the two market alternatives. On that basis we iteratively refined the design, incorporating visual aids and images to represent attributes and to facilitate comparison between the status quo and the alternative. Appendix Figure A6 presents the final format.

Experiment Design: Each respondent completed six choice tasks. The first task served as a practice round (round 0) to familiarise participants with the choice format and has been excluded from the analysis. The remaining five rounds (rounds 1–5) constitute our main analytical sample. In each choice task, respondents chose between two alternatives: (1) remain in their current location, or (2) relocate to a vending zone. Both alternatives were described by the same six attributes, whose values for the zone varied systematically across choice occasions, with the exception that the zone consistently offered legal rights to operate, the feature distinguishing formal from informal vending.

Vendors are told that the new marketplace is identical to their current one except in the attributes shown, and that storage capacity and transport costs are unchanged. This holds fixed the supply-side conditions that would otherwise confound valuations of the offered terms, so that differences across profiles operate through the attributes we vary. Appendix A6.3 repro-

duces the enumerator script.

Attribute Selection and Levels: The six attributes were selected based on formative qualitative interviews and existing literature on informal sector formalisation. Table 2 presents the full set of attributes and their possible values in the vending zone alternative. Attribute levels were determined through pilot testing to reflect realistic variation observed in actual vending markets while ensuring sufficient spread to identify preferences. Appendix A6 details how the scenarios for the experiment were selected.

Table 2: Possible Values of Attributes in the New Marketplace

Attributes	Levels
Monthly Rent (% of monthly sales)	0, 1, 2, 4, 8
Number of spots	same, 20, 40, 80, 160
Expected number of vendors relocating (% of vendors)	[0, 100]
Expected number of competitors (absolute change)	-10, -5, 0, +5, +10
Expected number of daily customers (% change)	-50, -20, 0, 20, 50 if current ∈ [10-60]
	-30, -15, 0, 15, 30 if current ∈ [61-120]
	-25, -10, 0, 10, 25 if current ∈ [121-170]
Distance from residence (absolute change)	-1, 0, +1, +3, +5

A key design feature is that attribute values for the alternative zone were defined *relative to each vendor's self-reported status quo*, elicited in the survey module preceding the choice tasks. For example, if a vendor reported a current commuting distance of 2 km and the design specified +1 km, the alternative was presented as a 3 km commute. All vendors therefore face the same experimental variation in relative terms while the absolute values are grounded in their actual situation. All pairwise attribute correlations in the design matrix are below 0.2, so individual attribute effects can be estimated without multicollinearity concerns (Appendix A6).

3.2 Discrete Choice Model

We estimate a McFadden conditional logit (McFadden 1974). Vendor i relocates in scenario t if the net utility gain from relocation exceeds zero:

$$U_{it}^{\text{relocate}} - U_{it}^{\text{stay}} = V_i(s) + \varepsilon_{it} > 0 \tag{1}$$

where ε_{it} is distributed Type I Extreme Value independently across i and t, and the deterministic component is:

$$\begin{aligned} V_i(s) = {} & \underbrace{\alpha}_{\text{vending zone}} + \underbrace{\delta_R \, \mathbf{1}\{\text{Rent}_i^{\%} > 1\}}_{\substack{\text{threshold} \\ \text{penalty}}} + \underbrace{\beta_1 (\text{Rent}_i^{\%} - 1)^+}_{\substack{\text{normalise WTP,} \\ \%\text{ of sales}}} + \underbrace{\beta_2 \Delta\text{Spots}_i}_{\text{zone size}} \\ & + \underbrace{\beta_3 \, s}_{\substack{\text{strategic} \\ \text{complementarity}}} + \underbrace{\beta_4 \Delta\text{Competition}_i}_{\text{rivalry}} + \underbrace{\beta_5 \Delta\text{Customers}_i}_{\text{demand}} \\ & + \underbrace{\delta_D \, \mathbf{1}\{\Delta\text{Distance}_i \neq 0\}}_{\substack{\text{fixed cost} \\ \text{of moving}}} + \underbrace{\beta_6 \Delta\text{Distance}_i}_{\substack{\text{commuting} \\ \text{cost}}} + \underbrace{\gamma' z_i}_{\substack{\text{vendor} \\ \text{characteristics}}} \end{aligned} \tag{2}$$

The intercept α is the alternative-specific constant on the vending zone. The coefficient β_3 on s identifies the strategic complementarity: how much the probability of relocation rises with the share of peers also relocating. z_i includes vendor-level controls for legal rights importance (a 1–5 scale), product category (vegetables, fruits, or cooked food and drinks, with cooked food as the omitted category), years in business, and risk aversion.

Two attributes enter non-linearly. Rent is assigned as a percentage of each vendor's own reported monthly sales, and δ_R and β_1 together allow it to carry no disutility up to a threshold of one per cent, a discrete penalty δ_R on crossing it, and a constant slope β_1 in the excess above. Distance enters as a one-off cost δ_D incurred whenever the commute differs from the status quo in either direction, plus a linear per-kilometre term β_6. Both forms are chosen rather than imposed: Appendix A7 reports the comparison against linear alternatives and free level dummies, and the profiling exercise that locates the rent threshold at one per cent. We refer to this specification as M5 in that appendix.

The implied choice probability is:

$$P(\text{relocate} \mid V_i(s)) = \frac{\exp(V_i(s))}{1 + \exp(V_i(s))} \tag{3}$$

When $V_i(s) = 0$, the vendor is indifferent: $P(\text{relocate}) = 0.5$. For the design attributes other than rent, willingness to pay is recovered by the ratio:

$$\text{WTP}_k = -\frac{\hat{\beta}_k}{\hat{\beta}_1} \tag{4}$$

measured in percentage points of monthly sales, since the numéraire β_1 is the rent slope above the threshold and rent was assigned as a share of each vendor's own sales. The recovered quantities are therefore rates on own sales rather than rupee amounts for a representative vendor, and rupee figures reported below are conversions at the median.
The value of the vending zone itself is not $-\hat{\alpha}/\hat{\beta}_1$, because the first percentage point of rent carries no disutility and crossing the threshold is costly in itself. It is instead the rent at which a zone matching the current spot on every other attribute leaves the vendor indifferent:

$$\text{WTP}_{\text{zone}} = 1 - \frac{\bar{\alpha} + \hat{\delta}_R}{\hat{\beta}_1}, \qquad \bar{\alpha} = \hat{\alpha} + \hat{\gamma}'\bar{z} \tag{5}$$

comprising the free first percentage point plus the intercept net of the threshold penalty, where $\bar{\alpha}$ evaluates the case variables at their sample shares so that the figure reflects the actual composition of vendor characteristics rather than an omitted category. Standard errors on both (4) and (5) are obtained by the delta method.

Because neither the threshold penalty nor the slope applies at a rent of zero, the value of the zone in the same units is instead

$$\text{Value}^{r=0}_{\text{zone}} = -\frac{\bar{\alpha}}{\hat{\beta}_1}, \tag{6}$$

which is the rent-equivalent value of the zone, the budget against which the attributes of the offer are drawn, as distinct from (5), the maximum rent the zone can carry.

The estimation sample uses rounds 1–3 only, excluding the practice round and post-treatment rounds 4–5 to avoid treatment contamination. Because vendors' repeated choices within the same marketplace may be correlated, through shared local information, similar zone options, or peer effects, standard errors are clustered at the marketplace level in our preferred specifications.

3.3 Information Experiment

The awareness data reported in Section 2 motivate the information experiment. At the start of the survey, we randomly assigned vendors to treatment and control groups within each marketplace. After completing round 3 of the choice tasks, treatment group vendors received detailed information about the Street Vendors Act (2014) and concrete evidence of enforcement in Chandigarh, where approximately 3,000 non-compliant vendors were displaced in late 2024 (the full treatment text is reproduced in Appendix A6.4). The information was delivered orally by enumerators to accommodate varying literacy levels. Control group vendors received no information and proceeded directly to round 4. All vendors then completed two further choice tasks (rounds 4–5) in the same format as previous rounds. Treatment and control groups are comparable on observable characteristics prior to information provision, with a marginal difference in weekly sales (Appendix Table A4).

To identify how enforcement information shifts the perceived value of the vending zone, we extend the indirect utility function to a difference-in-differences (DiD) specification. Rather than interacting treatment with each individual attribute, which would test whether information changes vendors' trade-offs over rent, distance, or spots, we allow treatment to shift only the intercept (baseline zone valuation), constraining attribute slopes to be common across treatment and control. This specification embodies the conceptually correct null: if the treatment conveys information about whether the zone will function as advertised, it should raise the overall value of the zone without changing the relative weights vendors place on its design attributes. The model is:

$$V_{ijt} = \underbrace{\alpha}_{\text{vending zone}} + \underbrace{\delta \cdot \text{treat}_i}_{\text{group difference}} + \underbrace{\mu \cdot \text{post}_t}_{\text{time trend}} + \underbrace{\lambda \cdot \text{treat}_i \cdot \text{post}_t}_{\text{treatment effect}} + \sum_k \underbrace{\beta_k x_{ijt}^k}_{\text{baseline effects}} + \underbrace{\gamma' z_i}_{\text{vendor characteristics}} \tag{7}$$

where $\text{post}_t \in \{0, 1\}$ indicates rounds 4–5 and $\text{treat}_i \in \{0, 1\}$ indicates treatment assignment. The coefficient of interest is λ: the differential change in the baseline zone valuation for treated vendors after information provision, with attribute slopes β_k held constant. δ captures any pre-existing difference in zone valuation between treatment and control groups (checked against balance); μ captures any time trend in zone valuation common to all vendors in the control group, absorbing learning effects, survey fatigue, or drift in response patterns unrelated to the treatment. The estimation sample for equation (7) includes all rounds 1–5 and is restricted to marginal vendors, for whom the treatment is operative. We report full sample estimates alongside. Standard errors remain clustered at the marketplace level.

4 Results

Table 3 presents conditional logit estimates across four specifications. Column (1) reports results for the full sample, without vendor-level controls; columns (2), (3) and (4) restrict the sample to marginal vendors, again without and with controls, with column (4) additionally clustering standard errors at the marketplace level to account for within-market correlation in choices. Estimates are stable across the three marginal vendor columns, so the results are not sensitive to the inclusion of controls or to the treatment of standard errors. The full sample column differs, most visibly in the one-off cost of moving, which attenuates towards zero once vendors whose choices do not vary are pooled in; Appendix A7 shows that this is dilution of a coefficient identified only off vendors who respond rather than a competing estimate. Column (4), marginal vendors with controls and clustered standard errors, is the main specification: it focuses on the policy-relevant group whose relocation decisions are sensitive to zone attributes, and clusters standard errors at the marketplace level, the level at which treatment and scenarios

were assigned. Clustering is not uniformly the more conservative choice: it inflates the standard errors on the design attributes and shrinks them on the case variables and the intercept. Appendix A7 reports the full variance ladder, including a pairs cluster bootstrap. All subsequent results, including the willingness to pay estimates in Table 4, are based on it. Section 4.3 characterises the marginal group itself and what its composition implies for which vendors a zone reaches.

Rent enters as an indicator for exceeding one per cent of monthly sales together with a linear term in the excess above that threshold, and distance as a one-off cost incurred whenever the commute changes at all together with a linear per-kilometre term. Appendix A7 reports the functional form comparison behind both choices and the profiling exercise that locates the rent threshold. Willingness to pay is measured in percentage points of monthly sales because the numéraire is the rent slope above the threshold. The design assigned rent as a percentage of each vendor's own reported sales, so the recovered parameters are rates on own sales rather than rupee amounts for a representative vendor. Rupee figures throughout are conversions at median monthly sales among marginal vendors, INR 65,000, the sample on which the estimates in column (4) are based. One percentage point of sales is therefore INR 650.

4.1 The Value of Formality

Vendors value what a zone offers. The alternative-specific constant prices vending zone membership, inclusive of legal rights, evaluated at the sample's actual composition of vendor characteristics. Expressed on the rent scale, a zone at the vendor's current location is worth 5.51 percentage points of monthly sales to marginal vendors. That figure is not a rent any vendor would pay: because rent carries no disutility below one per cent of sales and crossing that threshold is costly in itself, the maximum rent such a zone could actually carry is lower, at 2.94 percentage points of sales (INR 1,912), comprising the free first percentage point plus the share-weighted intercept net of the threshold penalty. The two answer different questions. The 5.51 points are the rent-equivalent value of the zone, the budget against which everything the policy asks of the vendor is drawn; the 2.94 points are the willingness to pay in the conventional sense, what a municipality could charge for designating an existing site. The gap between them is the price of crossing the threshold, and it exists only because rent is not linear through zero.

What vendors say they are buying is consistent with those magnitudes. Among those choosing the zone, 86% cite protection from eviction or harassment and 60% the reduction of stress and uncertainty, while expected footfall and amenities are cited less often (Figure 10, panel (b)). The value of formality here is largely insurance value, which is what the formalisation literature would lead one to expect, since registration raises neither productivity nor earnings for most firms and the welfare case rests instead on risk reduction and, for own-account entrepreneurs, on inclusion in social insurance (de Mel, McKenzie, and Woodruff 2013; Benhassine et al. 2018; Farias and Hsu Rocha 2026; Ulyssea 2020).

Stated attitudes do not predict choice. Figure 6 plots the predicted relocation probability against self-reported importance of legal rights for marginal vendors. The relationship is flat. Point estimates rise across the upper part of the scale, but no individual coefficient is distinguishable from zero and the confidence intervals span most of the plotted range (Table 3, column 4). This does not sit in tension with the valuation above. That figure rests on the alternative-specific constant, which prices the legal right as a feature every zone alternative carries. Figure 6 asks a different question: whether variation in what vendors say about the importance of legal rights predicts their choices, conditional on the right being on offer in every scenario. Formality can be worth a great deal on average while stated attitudes towards it explain nothing about who accepts it. At the policy-relevant margin, it is zone design rather

Table 3: Estimation Results: Conditional Logit Model

	Full Sample		Marginal Vendors					
	(1)		(2)		(3)		(4)	
$\mathbb{1}\{\text{Rent} > 1\%\}$	-0.344	***	-0.882	***	-0.889	***	-0.889	***
	(0.072)		(0.149)		(0.150)		(0.149)	
Rent above 1% (pp of sales)	-0.138	***	-0.247	***	-0.249	***	-0.249	***
	(0.021)		(0.042)		(0.042)		(0.042)	
Vendor Capacity (10s)	0.020	***	-0.007		-0.007		-0.007	
	(0.004)		(0.008)		(0.008)		(0.007)	
Relocating Vendors (%)	0.011	***	0.003	*	0.002		0.002	**
	(0.001)		(0.001)		(0.001)		(0.001)	
Competitors (10s)	-0.082	***	-0.051		-0.053		-0.053	
	(0.023)		(0.043)		(0.044)		(0.066)	
Daily Customers (10s)	0.054	***	0.102	***	0.099	***	0.099	***
	(0.017)		(0.035)		(0.035)		(0.035)	
$\mathbb{1}[\Delta\text{Commuting distance} \neq 0]$	-0.060		-0.707	***	-0.714	***	-0.714	***
	(0.068)		(0.143)		(0.143)		(0.149)	
Distance from Residence (kms)	-0.174	***	-0.329	***	-0.330	***	-0.330	***
	(0.018)		(0.038)		(0.038)		(0.043)	
Product Category								
Vegetables					0.015		0.015	
					(0.116)		(0.092)	
Fruits					-0.054		-0.054	
					(0.116)		(0.081)	
Legal Right Importance								
A little important					0.263		0.263	
					(0.541)		(0.440)	
Somewhat important					0.115		0.115	
					(0.550)		(0.427)	
Very important					0.460		0.460	
					(0.529)		(0.429)	
Extremely important					0.533		0.533	
					(0.550)		(0.453)	
Years in Business								
1 to 5 years					0.187		0.187	
					(0.212)		(0.149)	
5 to 10 years					0.073		0.073	
					(0.216)		(0.156)	
More than 10 years					0.115		0.115	
					(0.200)		(0.142)	
Risk Aversion					-0.004		-0.004	
					(0.028)		(0.019)	
Intercept	0.348	***	1.349	***	0.874		0.874	**
	(0.044)		(0.100)		(0.561)		(0.436)	
Number of observations	17532		4674		4674		4674	

Note: *** p<.01, ** p<.05, * p<.1. Column (1) is the full pre-treatment sample; columns (2) to (4) restrict to marginal vendors, those choosing to stay in one or two of the three scored tasks. Columns (3) and (4) add vendor characteristics. In column (4), the adopted specification, standard errors are clustered at the marketplace level.

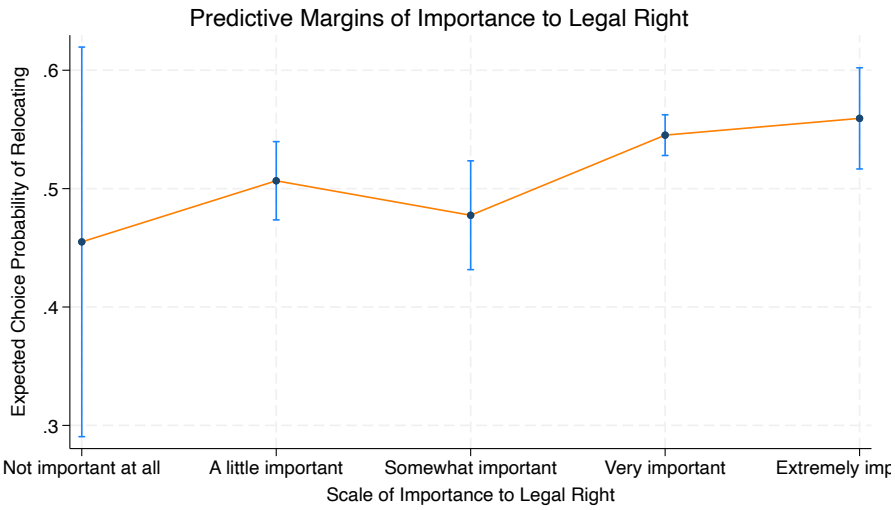
Table 4: Willingness to Pay

Vending Zone	2.942	***
	(0.386)	
Cost Of Moving	-2.869	***
	(0.694)	
Distance Per Km	-1.326	***
	(0.276)	
Marketplace Capacity	-0.0282	
	(0.028)	
Relocating Vendors	0.009	**
	(0.004)	
Competitors	-0.214	
	(0.266)	
Daily Customers	0.399	***
	(0.133)	
Number of observations	4674	

Note: *** p<.01, ** p<.05, * p<.1. WTP is measured in percentage points of monthly sales because the denominator is the rent slope above the one per cent threshold. Vending Zone is the maximum rent a vendor would accept for a zone matching their current spot on every other attribute: it comprises the first one per cent, which carries no disutility, plus the sample-share-weighted intercept net of the penalty for crossing the threshold. Cost of Moving is a one-off charge incurred whenever the commute changes at all, in either direction; Distance per km is the marginal cost on top of it. Relocating Vendors is per percentage point of the expected share of peers moving; Competitors and Daily Customers are per ten.

than professed enthusiasm for formality that converts marginal vendors into movers.

Figure 6: Choice Probability of Vending Zone Dependent on Importance Given to Legal Rights



4.2 The Cost of Moving

The largest single cost we estimate is incurred at the moment the commute changes, irrespective of by how much. A zone whose distance from the vendor's residence differs at all from the status quo carries a penalty of 2.87 percentage points of sales (INR 1,865), which is over half of the 5.51 points available at a rent of zero. That much of the value of the legal right is consumed before any distance is travelled and before any rent is charged. Beyond the one-off cost, each additional kilometre of commuting distance is worth a further 1.33 percentage points of sales

(INR 862), recovered by dividing the distance coefficient (−0.330) by the rent slope above the threshold (−0.249). The discrete term is not simple status-quo bias: estimated separately, the first kilometre costs roughly three times each subsequent one, and no comparable penalty attaches to moving away from the current rent or stall size (Appendix A7).

Rent itself is not linear in the amount charged. Below one per cent of monthly sales it carries no measurable disutility, so a modest rent is not what deters relocation and reductions inside that band buy nothing. Crossing the threshold carries a discrete penalty of 3.57 percentage points of sales (INR 2,321), equivalent to some 2.7 kilometres of commuting distance, and each further percentage point above it costs 1.00 point of sales by construction.

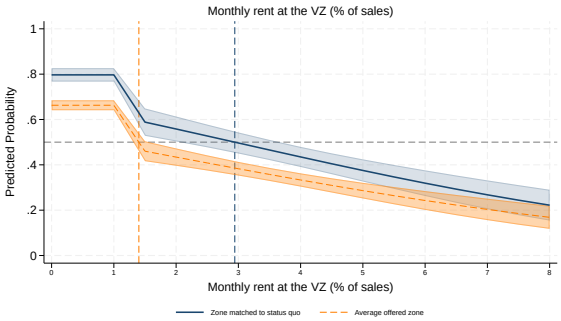
Two consequences follow from the size of the fixed cost relative to the budget it is drawn against. The first concerns what a relocated vendor can be charged. A move leaves 2.64 of the original 5.51 points, which is less than the 3.57 points that crossing the rent threshold costs, so the maximum rent a zone can support falls from 2.94 per cent of sales to one per cent on the fact of moving alone, whatever the distance involved. Relocation and a rent above the free band are close to mutually exclusive. The second concerns distance. Those 2.64 points are exhausted by a change in commute of 1.99 kilometres, so a zone whose commute exceeds the current one by about two kilometres is refused even when the spot is free. Because rent inside the free band costs nothing, that bound is the same whether the spot is free or charged at one per cent. Siting a zone somewhat closer recovers the per-kilometre component and never the fixed one.

Figure 7 plots how the predicted probability of relocation changes across the values of each attribute. The dashed curve holds the remaining attributes at the values actually drawn in the choice tasks, and so describes the average zone the experiment offered. The solid curve sets them to each vendor's own status quo, so that the zone differs from the current spot on the plotted attribute alone; this is the object priced in Table 4, and the rent panel accordingly crosses indifference at 2.94 per cent of sales. The vertical distance between the two curves is the value the average offered design gives away before charging any rent.

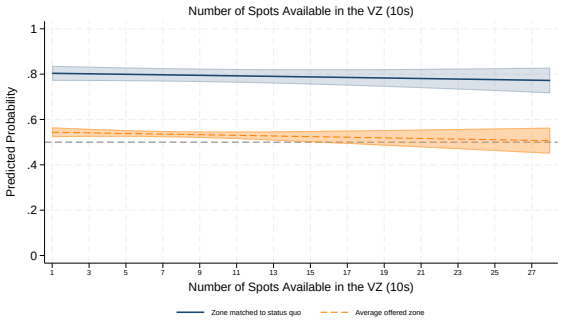
That gap is almost entirely commuting distance. The hypothetical zones presented in the choice tasks lay on average 1.19 km further from the vendor's residence than the current spot; these are the profiles our design generated rather than the zones PMC has built, whose distance from each vendor we do not measure. At the estimated parameters a zone at that average distance costs 4.45 percentage points of sales (INR 2,889): 2.87 points for changing the commute at all and a further 1.58 for the distance itself. The other four attributes add back only 0.39 percentage points of sales between them, almost all of it from an expected relocating share 41 percentage points above the status quo, worth 0.37 points (INR 240). Distance alone therefore consumes four fifths of the 5.51 points available. The consequence is not a proportional reduction in what a vendor will pay but a collapse to the free band: the value left in the average offered zone falls well short of the penalty for crossing the rent threshold. Because the design also included profiles that held the commute unchanged, and these avoid the one-off cost entirely, the average cost across profiles is smaller than the cost at the average distance, and at the rents the design actually drew the zone is still chosen with an unconditional predicted probability of 0.54. The design was not constructed to demonstrate any of this, since attribute levels were drawn to span a realistic range rather than to centre on the status quo. It is nonetheless the paper's argument appearing inside the experiment: siting consumes most of the value of formality before rent is charged.

Figure 7: Effects of Attributes on Choice Probability of Vending Zone

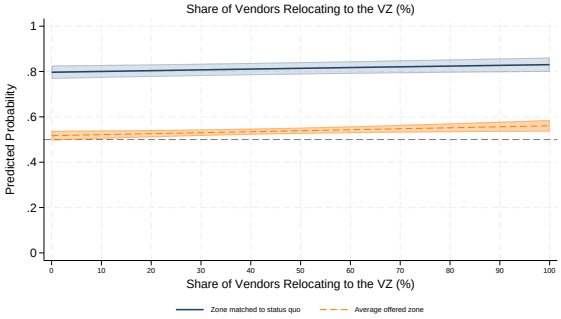
(a) Monthly Rent



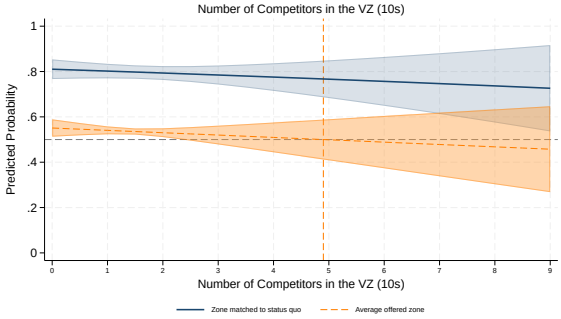
(b) Number of Spots in Vending Zone



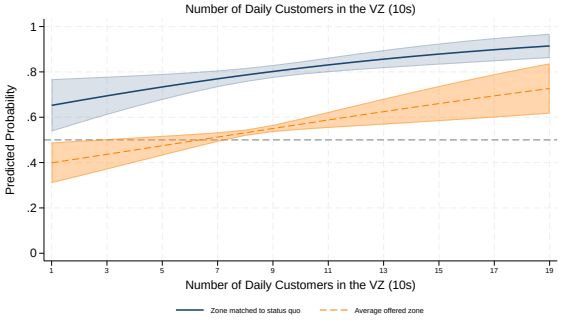
(c) Expected Share of Vendors Relocating



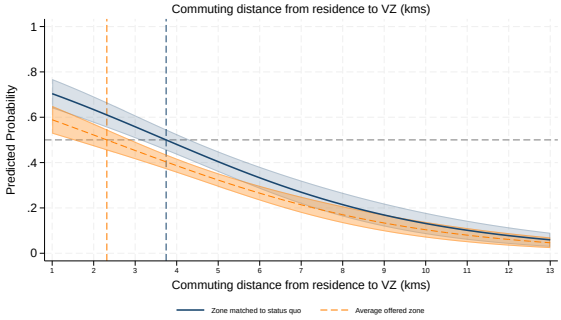
(d) Expected Number of Competitors



(e) Expected Number of Customers



(f) Distance from Residence



Note: All panels use column (4) of Table 3. The solid line sets all attributes other than the one plotted to the vendor's own status quo, so the zone differs from the current spot on that attribute alone. The dashed line leaves them at their realised design values. Sample includes marginal vendors with controls and standard errors clustered at the marketplace level.

4.3 Which Vendors a Zone Reaches

We classify vendors into three types based on their choices across rounds 1–3. *Always-movers* relocate in all three rounds regardless of the zone attributes offered; they are vendors for whom the vending zone is unambiguously preferred across all scenario draws. *Never-movers* never relocate across any of the three rounds; they cannot be induced to move by zone design changes. *Marginal vendors* exhibit variation in choices across scenarios; their relocation decisions are sensitive to zone attributes and represent the policy-relevant margin on which the estimates above are recovered. This classification uses only pre-treatment rounds to avoid contamination from the information experiment.

Figure 8 shows the distribution: 44% are always-movers, 30% never-movers, and 26% marginal vendors. Nearly three quarters of the sample therefore make the same choice in all three rounds, across the range of profiles the design drew, which raises the question of what separates them from the group whose choices respond.

Figure 8: Share of Vendors by Mover Status

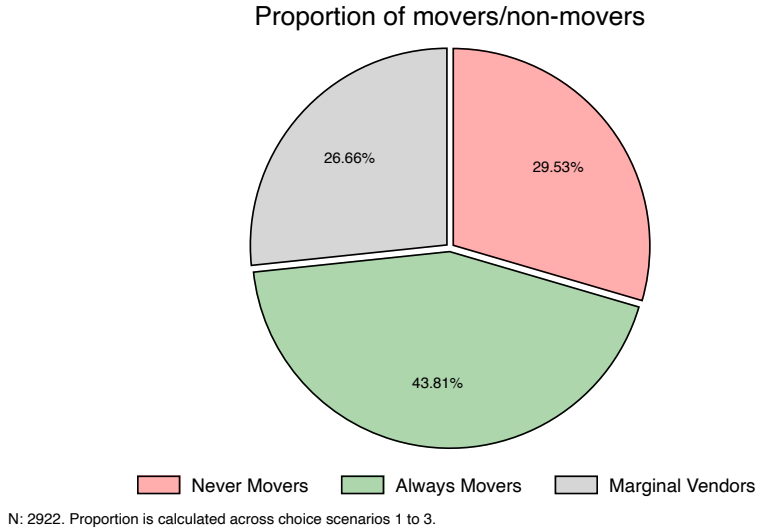
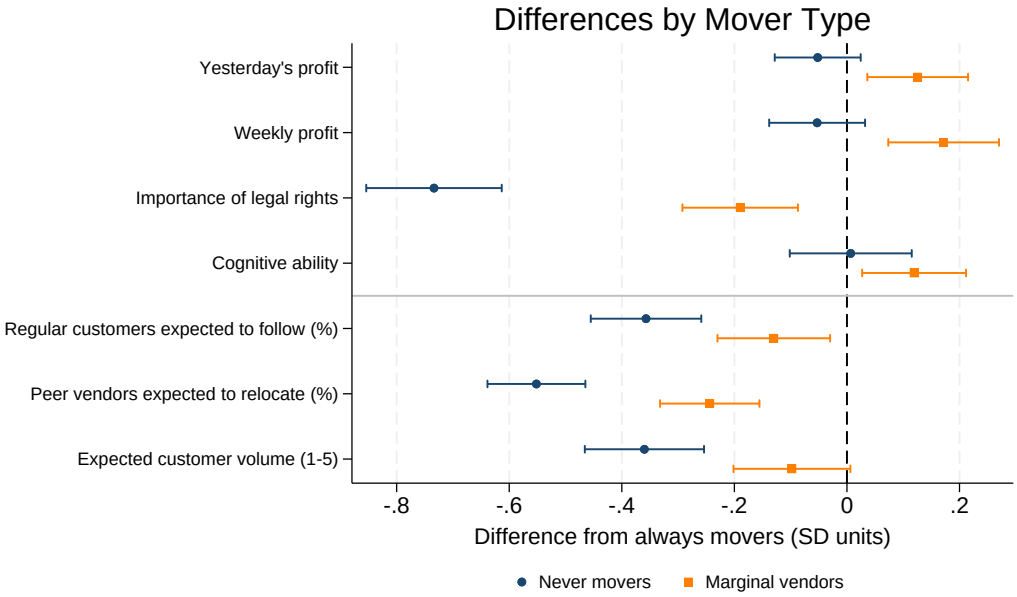


Figure 9 answers it. Each estimate comes from a within-marketplace regression of one standardised outcome on vendor type, with always-movers as the omitted category and marketplace fixed effects absorbed, so the differences are between vendors trading in the same market. The upper block reports characteristics of the vendor and their business; the lower block reports expectations about a hypothetical new vending area, elicited before any choice profile was shown. Reporting everything in standard deviations makes the two blocks comparable, which is what the pattern turns on. Table A3 reports the four characteristics in levels.

The two groups that do not respond to zone design differ from always-movers in entirely different places. Never-movers are indistinguishable from always-movers on both profit measures and on cognitive ability, but they value legal rights 0.73 standard deviations lower and are pessimistic on every belief margin, expecting fewer of their customers to follow, fewer of their peers to relocate, and lower customer volume at the destination, by between a third and just over half a standard deviation. What holds them in place is not what their current position is worth but what they take formality to be worth and what they expect a zone to deliver.

Marginal vendors show the opposite profile. They earn more than always-movers in the same marketplace, by 0.13 standard deviations on yesterday's profit and 0.17 on weekly profit, and

Figure 9: Differences between Mover Types



Note: Each row reports coefficients with 95% confidence intervals from a separate OLS regression of a standardized outcome on mover type, controlling for product category, ward and marketplace fixed effects. Always movers are the omitted category. Standard errors are clustered at the marketplace level. Profits are in INR. Importance of legal rights is measured on a 0–4 scale, from "not important at all" to "extremely important". Cognitive ability is the number of correct answers (0–3) to a math, a pattern recognition and a word recall question. The three expectation rows were elicited before the choice tasks. They cover the percentage of current regular customers expected to follow the vendor to the new marketplace, the percentage of peer vendors in the current marketplace expected to relocate, and expected customer volume in the new marketplace relative to the current location on a 1–5 scale (much fewer to much more). Vendors who could not predict customer volume (216) are excluded from that regression. $N = 2{,}922$ for the first four outcomes, 2,915 for regular customers, 2,895 for peer vendors, and 2,706 for customer volume.

score 0.12 standard deviations higher on cognitive ability, making them the highest-earning and highest-scoring group in the sample, with the profit difference concentrated in the upper tail.[2] Their beliefs sit closer to those of always-movers than to those of never-movers, and they value legal rights 0.19 standard deviations lower. What holds them in place is therefore the value of what they would be giving up. Place-based formalisation requires surrendering an incumbent location, so the more a vendor's current spot is worth, the more a move must deliver before it is worth accepting, and the vendors with most to lose are precisely those whose decisions turn on the terms of the offer rather than being settled in advance.

Because both non-moving groups value legal rights less than always-movers, what distinguishes the unconditional movers is an attitude towards formality rather than an endowment.[3] A zone would draw always-movers first, and would therefore open without the strongest traders in the marketplace, and plausibly without the customers they attract. The highest earners are reached only if the design is good enough, and whether they follow depends on the spatial and economic attributes of the zone rather than on how vendors feel about formality. A zone offering poor terms selects on a low opportunity cost of location, which is the inverse of what a policy aimed at productivity would want. The question is not whether these vendors are capable of formalising, but how much they must give up to do so.

Inattention does not account for the invariant responses. Only 4.2% of respondents failed the embedded attention check, and if satisficing drove the pattern it should concentrate among lower-scoring respondents, whereas never-movers, the group least willing to relocate on any terms, score no lower on cognitive ability than always-movers and are no poorer. That argument does not extend to always-movers, who score below marginal vendors on the same measure; Section 4.5 gives a separate reason for reading their responses as a genuine willingness to relocate.

The levels behind the belief differences bear on how costly relocation is, and not only on who bears the cost. Across the sample, only 14.9% of regular customers are expected to follow to a new site and 79% of vendors expect lower customer volume there (Figure A2). Because these were elicited before any profile was shown, the ordering across types cannot be a rationalisation of the choices made afterwards: the vendors least willing to relocate are those who expect to lose the most by leaving.

The stated reasons for vendors' choices make the same point in their own words. Figure 10 reports the reasons elicited after the final choice task, for staying in panel (a) and for relocating in panel (b). Among marginal vendors choosing to stay, 60.2% cite high rent and 44.2% cite distance from home. Never-movers place relatively greater weight on features of the incumbent spot itself: an established customer base, the time required to rebuild it, and attachment to the current area. Coordination concerns are comparatively uncommon in both groups.[4] For firms operating without premises, the price of formality is inherently spatial: it requires exchanging

[2] These are differences in conditional means. Median weekly profits are INR 3,500 for both marginal vendors and the full sample, so the premium reflects a heavier right tail among marginal vendors rather than an upward shift of the entire distribution.

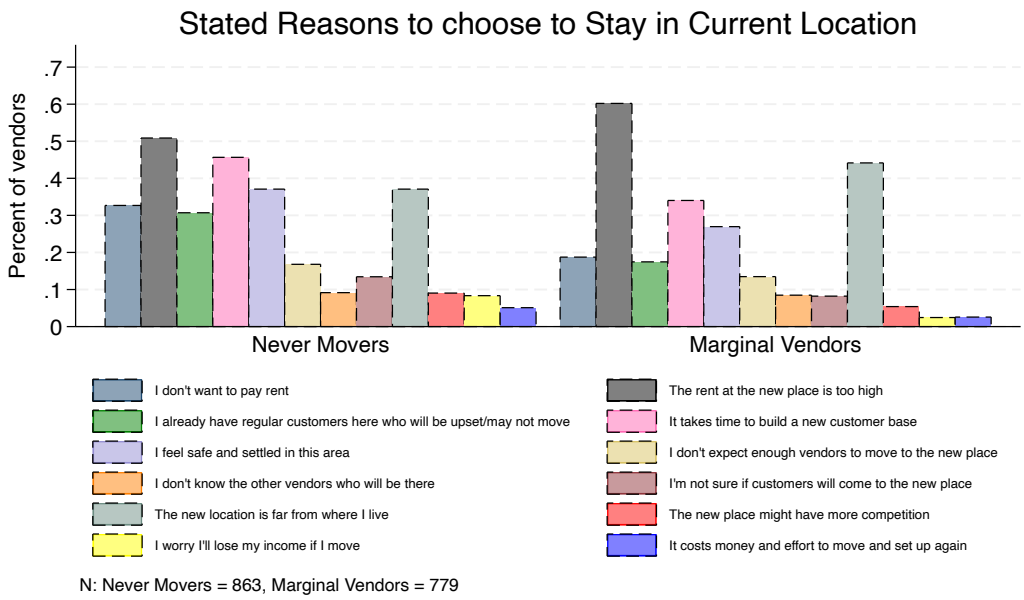
[3] This between-type difference is a different object from the within-group variation in Section 4.1, where stated importance of legal rights does not predict choice among marginal vendors. The comparison here is across types on an attitude; there it is within the marginal group, conditional on the legal right being attached to every zone alternative.

[4] Mover types do not differ in reported ties within the marketplace either. Vendors were asked how many others in their market they drink tea with, could ask to mind their stall, share business information with, could borrow from, would lend to, and are related to. Regressing each count, as a share of the marketplace, on mover type within marketplaces gives estimates that are individually statistically insignificant. These counts cover relations with other vendors in the current marketplace, so they do not speak to attachment to customers, to the location itself, or to ties elsewhere in the city. Appendix A5 reports the estimates.

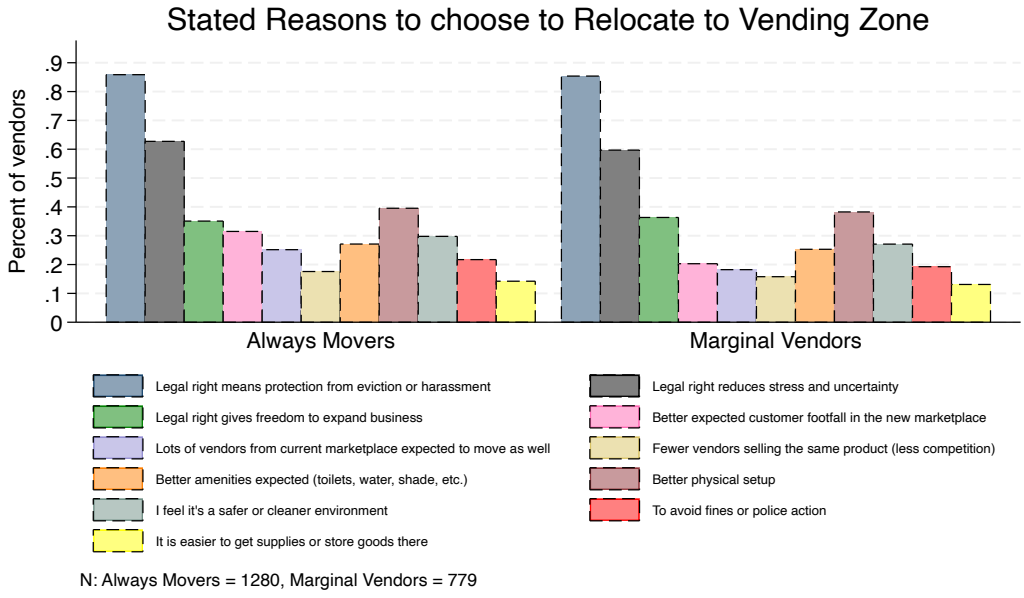
an established location for a new one.

Figure 10: Decision to Stay or Relocate by Mover Type

(a) Stay



(b) Relocate



Note: Reasons are elicited once, after the final choice task. Panel (a) reports the share of never-movers and marginal vendors citing each reason for choosing to stay; Panel (b) reports the share of always-movers and marginal vendors citing each reason for choosing the vending zone.

4.4 What Does Not Explain Empty Zones

Three of the four explanations commonly advanced for unoccupied vending zones account for much less than the spatial cost at the levels a zone would plausibly offer, and one further design response can be ruled out alongside them. It is their joint weakness, rather than the size of the distance estimate on its own, that makes siting the binding constraint.

Coordination effects: present but small. The peer relocation share s enters with a positive coefficient in every specification, significant in the full sample and at the 5% level among marginal vendors once standard errors are clustered (Table 3). Vendors are therefore more likely to relocate when more of their peers do. The magnitude is small. The implied willingness to pay for a ten percentage point increase in s is 0.09 percentage points of sales (INR 59) for marginal vendors, about 7% of the willingness to pay for a one kilometre reduction in distance and around 3% of the one-off cost of moving. Panel (c) of Figure 7 shows the slope directly, and moving from no coordination to full coordination ($s = 0$ to $s = 100$) is worth about as much as 0.68 km of commuting distance, which is a comparison at the margin only, since no reduction in distance is available to a relocating vendor without the one-off cost being incurred first. Because the design varies s across its entire range, this bounds the mechanism from above: there is no larger coordination effect available at any peer share a policy could credibly promise. Appendix A7 bounds it from above on a second ground, that the assigned share was constructed around each vendor's own elicited belief, so the coefficient is not identified by the experiment alone.

The contrast with the belief data in Figure A2 shows why that experimental variation is necessary. Expected peer relocation is strongly associated with mover type, with never-movers expecting 29.4% of their peers to move against a sample mean of 40.7%. Read cross-sectionally, that association would place coordination at the centre of the problem. But the types differ on several margins at once, and never-movers also value legal rights substantially less than always-movers (Section 4.3). The types differ in what they expect of their peers but not in the number of vendors they report knowing in their own market (Appendix A5). Pessimism about peers travels with a lower valuation of the zone itself, so the observational association confounds the effect of peer behaviour with whatever else separates the types, which is the simultaneity problem set out in Section 3 appearing in the data. When s is assigned rather than believed, the effect is an order of magnitude smaller than the spatial and pricing margins. Zones are not empty because vendors are locked in a low relocation equilibrium; they are empty because the zones on offer are too far away and too expensive.

Zone size. The number of spots offered in the zone does not attract vendors. For marginal vendors the estimated willingness to pay is -0.03 percentage points of sales per ten additional spots (INR 18) with a standard error of 0.028 (Table 4), so that moving across the full design range from 20 to 160 spots is worth around -0.40 points (INR 256) and cannot be distinguished from zero. The point estimate is negative in every marginal vendor specification we estimate, and negative and significant at the ten per cent level in the five round specification of Table 5, though positive in the full sample. One reading is that, conditional on the expected number of competitors and the expected share of peers relocating, additional spots represent room for entrants who are not yet present, which a vendor with an established position has no reason to value. The estimate is small enough that we do not lean on that interpretation. What it does rule out is the simplest policy response to an empty zone, which is to build a larger one. This attribute is the number of spots, not the floor area of each one, which the choice tasks hold fixed by instructing vendors to assume an unchanged scale of operation; Section 4.5 treats spot dimensions as a feature of the Bihar rules rather than as an estimated quantity.

Enforcement information. Table 5 presents conditional logit estimates of equation (7), restricted to marginal vendors. The Treat×Post coefficient is 0.315 ($p < 0.05$), indicating that treated vendors assign higher baseline value to the vending zone after receiving enforcement information. Translating this into willingness to pay against the rent slope estimated in the same model, the information treatment raises the perceived value of the vending zone by 1.75 percentage points of monthly sales (INR 1,138). For context, this is meaningfully positive but smaller than the value of closing a one-kilometre distance gap, which the same model prices at

2.16 points of sales (INR 1,405), and smaller still than the one-off cost of changing the commute at all, at 2.51 points (INR 1,629). The five round sample estimates a flatter rent slope and steeper distance terms than the three round sample of Table 3, so these ratios are reported within model and are not directly comparable with the magnitudes in Table 4.

Given that only 1.3% of vendors were aware of the SVA at baseline (Figure 5), this result is consistent with the information channel: most vendors simply did not know the policy existed or was being enforced elsewhere, and learning this raises the value they place on the vending zone. Because our treatment combines general policy information with a specific example of enforcement, we cannot separately identify whether the effect operates through basic awareness or through the credibility conferred by a concrete example; the extremely low baseline awareness in our sample makes this distinction difficult to isolate. That same low baseline means the estimate bounds what information provision can achieve rather than describing the return to a marginal campaign. What the results do establish is that information can modestly raise the perceived value of vending zones, but cannot substitute for addressing the underlying rent and distance frictions.

Rent below the threshold. The fourth explanation, that rents are unaffordable, is not supported at the levels a zone is likely to charge, for the reasons set out in Section 4.2. The shape of the rent response reflects the counterfactual: vendors pay nothing for their current pitch and only 9% make any informal payment, so a zone rent is a new obligation rather than a substitute for one already borne. What is expensive is crossing the threshold, and the design question is therefore not how low to set the rent but whether to cross that threshold at all.

4.5 Space Constraint and the Formalisation Gap

The one dimension of zone design our experiment cannot price is the space allocated to each vendor, because the choice tasks hold operating scale fixed by construction. Under the Bihar implementation of the Act each spot measures 6 by 4 or 5 by 3 feet. In our sample 85% of vendors operate from a stall or cart rather than a floor display (Table 1), and many sell perishable produce or cooked food requiring storage, preparation area, and customer-facing display. We do not measure current footprints, so the comparison rests on the stated spot dimensions alongside the observed distribution of stall types rather than on a direct estimate.

This offers an explanation for the divergence between stated willingness to relocate (44% always-movers in our experiment) and near-zero actual take-up. Our stated preferences experiment tells vendors the zone differs from their current spot only on the attributes shown, and that storage capacity is unchanged; operating scale is therefore held fixed by construction, and the space constraint is bracketed. Always-movers are therefore vendors who would genuinely relocate if a zone could accommodate them at their current scale. The fact that actual zones cannot do so means that even the most relocation-willing vendors face a structural impediment that rent subsidies, information campaigns, or coordination mechanisms cannot resolve without first addressing the allocated space per vendor. This is a second spatial margin alongside the siting of the zone.

5 Conclusion

Place-based formalisation fails on the terms of the offer rather than on the value of what is offered. The legal right itself is not what vendors are rejecting: offered a zone at their current location and charged no rent, marginal vendors value it at 5.51 per cent of monthly sales, a fifth of the sample already holds a certificate of vending, and 86% of those choosing a zone cite

Table 5: Information Experiment Estimation Results: Conditional Logit Model

	Full Sample (1)		Marginal Vendors (2)	
$\mathbb{1}\{\text{Rent}^{\%} > 1\}$	-0.377	***	-1.015	***
	(0.061)		(0.134)	
Rent above 1% (pp of sales)	-0.133	***	-0.180	***
	(0.016)		(0.030)	
Vendor Capacity (10s)	0.016	***	-0.010	*
	(0.005)		(0.006)	
Relocating Vendors (%)	0.010	***	0.003	***
	(0.001)		(0.001)	
Competitors (10s)	-0.068	***	-0.094	
	(0.024)		(0.080)	
Daily Customers (10s)	0.043	***	0.099	***
	(0.013)		(0.023)	
$\mathbb{1}[\Delta\text{Commuting distance} \neq 0]$	-0.038		-0.451	***
	(0.058)		(0.130)	
Distance from Residence (kms)	-0.181	***	-0.389	***
	(0.019)		(0.036)	
Treatment Group	0.004		0.012	
	(0.068)		(0.068)	
Post Treatment	-0.107	***	-0.028	
	(0.040)		(0.085)	
Treat x Post	0.178	***	0.315	**
	(0.058)		(0.126)	
Product Category				
Vegetables	-0.044		-0.102	
	(0.088)		(0.103)	
Fruits	0.113		-0.153	*
	(0.089)		(0.083)	
Legal Right Importance				
A little important	2.298	***	0.286	
	(0.389)		(0.529)	
Somewhat important	2.279	***	0.230	
	(0.383)		(0.536)	
Very important	3.179	***	0.631	
	(0.383)		(0.521)	
Extremely important	3.983	***	0.917	*
	(0.390)		(0.531)	
Years in Business				
1 to 5 years	0.098		0.131	
	(0.135)		(0.134)	
5 to 10 years	0.082		0.088	
	(0.158)		(0.151)	
More than 10 years	0.225		0.122	
	(0.141)		(0.133)	
Risk Aversion	0.004		-0.017	
	(0.018)		(0.020)	
Intercept	-2.760	***	0.728	
	(0.390)		(0.531)	
Number of observations	29220		7790	

Note: *** p<.01, ** p<.05, * p<.1. Estimated on all scored rounds, the practice round excluded. Treatment shifts the vending-zone intercept only, with attribute slopes held common across arms and periods. Treat x Post is the differential change in the baseline zone valuation for treated vendors after information provision. Standard errors are clustered at the marketplace level.

protection from eviction or harassment as the reason. What consumes that value is relocation, and it does so discontinuously. A vendor forfeits 2.87 percentage points of sales the moment the commute changes at all, over half the available value, before any distance is travelled and before any rent is charged; each further kilometre costs 1.33 points, and the 2.64 points that survive the move are exhausted by about two kilometres. What survives also cannot carry a rent above one per cent of sales, against the 2.94 per cent the same zone could command if designated in place. The three other explanations for empty zones account for much less. Coordination failure, the account most often advanced, is real but small: moving from no coordination to full coordination is worth about as much as 0.68 km of commuting distance. Building larger zones is not the remedy either, since the number of spots a zone offers has no detectable effect on relocation and among marginal vendors the point estimate is negative. And information, though it raises the perceived value of a zone, leaves the distance and rent trade-offs untouched. What has to change is whether the vendor has to move at all, the distance to the zone if they do, the rent charged, and the floor area of the spot itself.

5.1 Implications for design

Three of those four margins are priced by our estimates and the last is not. The spot dimensions fixed by the Bihar rules are smaller than the space most vendors currently occupy to run their operations, so relocation would require a reduction in operating scale that our choice tasks hold constant by construction (Section 4.5). Expanding the space allocated per vendor is therefore a precondition for translating stated willingness into actual take-up, including among the always-mover population, 44% of the sample, who would relocate under almost any reasonable zone design.

The policy-relevant margin beyond always-movers is marginal vendors, who can afford rent but are sensitive to where a zone sits and what it charges. Our estimates give a set of exchange rates for reaching them. The first is not an exchange rate at all: because the cost of moving is incurred in full as soon as the commute changes, siting a zone closer recovers only the per-kilometre component and never the 2.87 points of sales that changing the commute costs in itself. Where an existing vending site can itself be designated as a zone, the legal right is delivered without triggering that cost, and this is close to the only option our estimates support rather than merely the cheapest. It is also the option the municipality has the least room to choose. The sites vendors occupy are the sites whose congestion the policy exists to relieve, and nine in ten vendors in our sample trade on main roads, so in-situ designation will be feasible in some marketplaces and not others. Our estimates price what displacement costs the vendor; they do not price what leaving vendors in place costs the city, and the comparison that would settle the choice at any particular site requires both.

Where displacement is unavoidable, 1.33 percentage points of sales per kilometre (INR 862) is the rate at which distance has to be bought back on top of the one-off cost, through closer siting or through lower rent. The scope for buying it back is narrow. A relocated vendor can carry at most one per cent of sales in rent, against 2.94 per cent for the same zone designated in place, so displacement costs the municipality something like 1.94 points of monthly rent per vendor (INR 1,262) in every case where the vendor accepts at all. Beyond about two kilometres nothing is left to trade: a zone at that distance is refused even when the spot is free.

Rent policy inside the free band is a separate matter. Vendors do not respond to rent below one per cent of monthly sales, so a charge at or under that level costs nothing in take-up, and reductions inside the band buy nothing in return. What is expensive is crossing the threshold, which carries a discrete penalty worth 3.57 points of sales, some 2.7 kilometres of commuting distance. The design question is not how low to set the rent but whether to cross that line at all.

Two further levers do less than is often supposed. Information provision raises the perceived value of a zone by 1.75 percentage points of monthly sales (INR 1,138), but because baseline awareness is 1.3% that figure bounds what information can achieve rather than describing the return to a marginal campaign, and it leaves the rent and distance trade-offs untouched. Enforcement is the instrument this literature identifies as most effective at inducing formalisation (Andrade, Bruhn, and McKenzie 2016; Ulyssea 2020), and it carries a corollary here. Raising the cost of informality while zone spots remain too small to house existing operations would not move these vendors into the formal economy; it would make their informality more expensive without opening a viable alternative. Singapore, the one case we found in which the empty zone equilibrium was decisively broken, did exactly this, withdrawing street hawker licences rather than improving the zones (Appendix A1), and the cost of any mismatch between what vendors needed and what the zones offered fell on the vendors.

Taken together, the estimates locate the failure of these programmes in a constraint rather than in a mistake. Zones sit empty not because the state has failed to offer formality or because vendors do not want it, but because the only instrument available for conferring legal status on a firm without premises requires moving it, and moving it consumes most of what the legal status is worth. The state is not absent here, and the firms are not myopic. What binds is that the site a vendor is most willing to be formalised in is the site the city is least willing to leave them in.

5.2 Transition costs and the long run

Our estimates price the decision to relocate at a point in time, and that decision bundles two things a policymaker would want to separate: the cost of moving and the cost of being somewhere else. Our specification separates them statistically, but not by their duration, and part of the one-off cost is plausibly transitional. Among vendors choosing to stay, 45.7% of never-movers and 34.0% of marginal vendors cite the time needed to rebuild a customer base, and smaller shares cite the money and effort of setting up again (Figure 10). Such costs are by construction temporary, so part of the 2.87 points of sales we attribute to changing the commute may be the disruption of moving rather than the standing disadvantage of the destination. If so, the short-run cost of relocation exceeds its long-run cost, and a design that looks unattractive on our estimates could still leave vendors better off once adjustment is complete. Several channels would push that way: secure tenure lengthens the horizon over which it is worth investing in a spot (Harari and Wong 2026), recognition carries eligibility for the collateral-free lending under PM SVANidhi, destinations need not stay as they begin (Franklin 2025; Agness and Getahun 2024), and a lawful spot removes a recurring source of loss rather than a one-off one. The evidence also cuts the other way, since relocated households in Mumbai were no better off than those who stayed (Barnhardt, Field, and Pande 2017) and formalisation raises neither productivity nor earnings for most firms (de Mel, McKenzie, and Woodruff 2013; Farias and Hsu Rocha 2026). We should be clear that this is speculation: we observe the choice and not what follows from it.

What does follow from our estimates, whichever way that evidence eventually settles, is an asymmetry worth taking seriously in design. The transition cost is borne by the vendor with certainty and immediately; the long-run gain is uncertain and deferred, and for a firm with the profit levels reported in Table 1 that asymmetry is not a matter of patience but of whether the intervening months are survivable. This is an argument for compensating the move rather than assuming it will pay for itself, whether through a rent holiday over the period a customer base is rebuilt or through the working capital that PM SVANidhi already provides.

5.3 Limitations

Several limitations bear on how these estimates should be read. They are recovered from stated rather than revealed choices. The design personalises every attribute to the vendor's verified circumstances, and vendors were told at the outset that their responses would inform PMC policy, but we cannot rule out that hypothetical choices overstate the level of willingness to relocate.

Because the choice tasks hold operating scale constant, the space allocated per vendor is bracketed rather than priced. We can say that the Bihar spot dimensions are smaller than the operations most vendors run and that this is consistent with the observed pattern of take-up, but not what a vendor would pay to avoid shrinking, nor how that cost compares with the distance and rent margins we do estimate. The account we give of the gap between stated willingness and observed take-up is therefore the most plausible one available to us rather than an estimated result, and it rests on the stated spot dimensions alongside the observed distribution of stall types rather than on measured footprints. More generally, six attributes do not exhaust the design of a vending zone. The duration of tenure security, the quality of enforcement inside the zone, infrastructure, and eligibility for complementary schemes such as PM SVANidhi are all held constant, so the estimates price the attributes we vary conditional on the rest of the offer being as vendors imagine it.

A related limit applies to the coordination estimate, which is identified from an anonymous share of relocating peers. Reported ties within a marketplace are few relative to its size (Appendix A5), and the design does not distinguish a vendor's response to that share from their response to knowing which particular vendors were moving.

Finally, we price the relocation decision and not its consequences. The experiment says nothing about whether vendors who relocate earn more, are evicted less, or are better off, so the welfare case for these programmes cannot be settled from preferences alone. Nor do we measure the congestion costs that motivate relocation in the first place, so our estimates bound what displacement costs the vendor without establishing when it is nonetheless warranted. Our setting is a single city, and while the instrument is common across the developing world (Appendix A1), the spot dimensions, rent levels and commuter-led demand structure in Patna are specific to it. The stated reasons in Figure 10 were elicited after the final choice round and are therefore post-treatment for vendors in the information arm, though the beliefs in Figure A2, elicited before the choice tasks, are not.

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Appendix

A1 Relocation Policies Around the World

Figure A1: News Articles from around the world



The formalisation model we study in Patna, in which vendors are offered legally protected spots in a designated zone in exchange for relocating, is not an idiosyncratic feature of Indian urban policy. Table A1 documents 54 policy instruments in 27 countries across six regions, all but one dated between 2001 and 2026; the exception is Singapore's hawker resettlement programme of the 1970s and 1980s, the historical template from which much of the contemporary model derives. Just over 70% of the entries involve designating or constructing a physical site to which vendors are expected to move, which is the margin our experiment evaluates. The approach spans low-income settings with very large informal sectors (Tanzania, Uganda, Bangladesh), middle-income settings undergoing rapid redevelopment (Indonesia, Vietnam, Peru, Egypt), and high-income ones where vending is a small but politically salient share of employment (Singapore, the United States), under governments of every stripe. It is also durable: R. Bromley (2000) reviewed the same family of instruments a quarter of a century ago, and the design has changed little since.

Table A1: Street vendor formalisation, relocation and vending zone policies

Country and city	Policy or programme	Type
Panel A: South Asia		
India	National Policy on Urban Street Vendors (2004, rev. 2009)	L
India	Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act (2014)	L, Z, R
India	PM SVANidhi micro-credit scheme (2020)	R
India, Bhubaneswar	Participatory vending zone regularisation (from 2007)	Z, R
Bangladesh, Dhaka	City Hawker Management Policy (2026)	L, Z, R
Bangladesh, Dhaka	DNCC temporary markets on six open fields (2026)	Z
Nepal, Kathmandu	KMC space allocation proposals and clearance (2010s)	E
Panel B: Southeast and East Asia		

Continued on next page

Table A1, continued

Country and city	Policy or programme	Type
Thailand, Bangkok	BMA vending ordinance (2002) and Designated Areas Directive (2005)	L, Z, R
Thailand, Bangkok	Return the Footpath to Pedestrians (from 2014) (Boonjubun 2017)	E, Z
Thailand, Bangkok	Revised BMA rules and hawker centre plan (2020s)	Z, R
Indonesia, Surakarta	Notoharjo and Panggung Rejo relocation (2006); Regulation No. 3 (2008) (Permatasari et al. 2014)	Z
Indonesia, Yogyakarta	Malioboro to Teras Malioboro relocation (2022)	Z
Indonesia, Jakarta	Relocation to designated lots (from 2015)	Z, E
Vietnam, Hanoi	Vending ban, 62 streets and 48 public spaces (2008) (Eidse, Turner, and Oswin 2016)	E
Vietnam, Ho Chi Minh City	Civilized Streets (2013 to 2015); sidewalk reclamation (2017)	E
Vietnam, Ho Chi Minh City	Nguyen Van Chiem and Bach Tung Diep food zones (2017)	Z, R
China	Street stall economy, State Council measures (2020) (Song 2020)	L, Z
China, Shenzhen	Designated vending areas (2023)	Z
Singapore	Hawker resettlement programme (1970s to 1986)	Z, R
Singapore	Hawker Centres Upgrading Programme (2001); SEHC model (2015)	Z
Philippines, Quezon City	Revised Market Code, Ordinance SP-2459 (2015)	L, Z, R
Philippines, Manila	Ordinance No. 8349 on designated vending areas	L, Z
Philippines	House Bill 10724, proposed national framework (2024)	L, Z, R
Panel C: Sub-Saharan Africa		
South Africa, Durban	Informal Economy Policy and Warwick Junction upgrade	L, Z
South Africa, Johannesburg	Operation Clean Sweep (2013) (Bénit-Gbaffou 2016)	E, Z
Tanzania	Vendor recognition (2016); Wamachinga ID cards (2018) (Msoka and Makundi 2023)	R
Tanzania, Dar es Salaam	Machinga Complex and Kariakoo pitch allocation (from 2017)	Z
Rwanda, Kigali	Vending ban and mini-market programme (2017, 2022) (Gisanabagabo and Masengesho 2024)	E, Z
Uganda, Kampala	KCCA gazetted markets directive (2022)	E, Z
Kenya, Nairobi	Pop-Up Markets and Street Vending Act (2019)	L, Z, R
Kenya, Nairobi	Mwariro and Karandini market relocation (2017 to 2019)	Z
Ghana, Accra	AMA decongestion by-laws (2011) and clearance exercises (Crentsil and Owusu 2018; Steel, Ujoranyi, and Owusu 2014)	E, Z
Zimbabwe	Operation Murambatsvina (2005) (Musoni 2010)	E, Z
Zimbabwe, Harare	Designated vending sites directive (2015)	E, Z
Liberia, Monrovia	Memorandum of Understanding with the City Corporation (2018)	L, R
Senegal, Dakar	Petersen Market eviction and PASI negotiations (2023)	E

Continued on next page

Table A1, continued

Country and city	Policy or programme	Type
Panel D: Middle East and North Africa		
Egypt, Cairo	Relocation of downtown vendors to Torgoman (2014)	E, Z
Egypt, Cairo	Six alternative public markets, incl. El Tunisi, El Turjuman and Ahmed Helmy (Mohammed Darwish 2023)	Z
Egypt, Cairo	Al-Tounsi Market relocation, UNESCO management plan (2010 to 2014)	Z
Panel E: Latin America		
Mexico, Mexico City	Programa Integral de Rehabilitacion del Centro Historico (2007) (Crossa 2009)	Z, E
Peru, Lima	La Parada closure and Santa Anita relocation (2012) (Coletto, Jaber, and Vanhellemont 2021)	Z, E
Colombia, Bogota	IPES REDEP kiosk network (Donovan 2008)	Z, R
Colombia, Bogota	Politica Publica de Vendedores Informales 2023 to 2035; RIVI register	L, R
Colombia, Bogota	Public space decree, confianza legitima protocol (2026)	L, Z
Ecuador, Quito	Relocation to eleven Centros Comerciales de Ahorro (2003) (Middleton 2003; R. D. F. Bromley 1998)	Z
Brazil	Microempreendedor Individual, Complementary Law 128 (2008)	L, R
Brazil, Sao Paulo	To Legal, Termo de Permissao de Uso for ambulantes	R
Chile, Santiago	Ordenanza No. 127 on comercio estacionado (rev. 2026)	L, R
Paraguay, Ciudad del Este	Municipal formalisation project	Z, R
Panel F: North America		
United States, California	Safe Sidewalk Vending Act, SB 946 (2018)	L, R
United States, Los Angeles	City and County Sidewalk Vending Ordinances (from 2018)	L, R
United States, California	SB 972, retail food code reform (2022)	L, R
United States, New York City	Local Law 18, permit expansion and enforcement office (2021)	L, R
United States, New York City	Advisory Board recommendations on designated spaces (2022)	Z

Notes: Non-exhaustive. Street vending is regulated mainly at municipal level, so the full universe runs to thousands of by-laws, most undocumented in academic or English-language sources. This is a curated sample weighted towards cases with an identifiable relocation or zone-creation component and a citable source. Academic treatments are cited where they exist; otherwise the policy name links to its primary source. Types: L, legislation or policy framework; Z, designation or construction of vending zones or market facilities; R, registration or licensing of vendors; E, clearance paired with an offer of an alternative site. Entries may carry more than one code. Spanish and Portuguese titles are given without diacritics.

A subset of these cases matches the Patna setting closely, in that a purpose-built facility was constructed and then under-occupied for a documented reason. Mohammed Darwish (2023) evaluates six alternative public markets in Cairo and identify unprofitable location as the crucial factor behind vendor rejection, alongside a failure to accommodate vendors' operating require-

ments in the design of the stalls. In Surakarta, the Notoharjo market was initially well attended but drew persistent complaints of insufficient footfall as its facilities deteriorated (Permatasari et al. 2014). In Lima, larger wholesalers moved to the new Santa Anita facility while smaller traders, for whom the rents were unaffordable, stayed put (Coletto, Jaber, and Vanhellemont 2021). In Kampala, a survey conducted after the 2022 directive to trade only from gazetted markets found average daily earnings of UGX 50,000 among street traders against UGX 7,000 among those trading from the zoned markets, with 85% of relocated traders reporting lower income (Gateway Research Centre 2022). In Nairobi, the Mwariro market was built with roughly 350 stalls against some 7,000 stalls whose traders were to be moved out of the central business district (PesaCheck, 2018), and stalls there remained unoccupied years after completion (*Daily Nation*, 2024). These map onto the three constraints our estimates identify: distance and foregone footfall, rent, and allocated space. The qualitative literature has recognised the pattern, with R. D. F. Bromley and Mackie (2009) showing how stall allocation rules and pricing in new Latin American municipal markets excluded the poorest traders, and Roever and Skinner (2016) reporting vendor complaints about the viability of relocation sites in Lima and Mexico City. What it does not do is quantify how much of the rejection is attributable to distance, to rent, or to the spot itself, since these move together across cases and cannot be separated with observational data. That decomposition is our contribution.

Singapore is the instructive contrast, being one of the few documented cases in which the empty zone equilibrium was broken. In 1975, one in six stalls in the new hawker centres sat empty while some 15,000 hawkers still traded on the street, and the reasons recorded for their reluctance were that street spots carried no rent, that hawkers had established clienteles they feared losing, and that a street stall could be enlarged simply by adding tables and chairs (National Archives of Singapore 2026). Those are the three margins our estimates identify, half a century earlier and on another continent. The equilibrium was reversed not by improving the zones but by removing the alternative: licensed hawkers had to occupy their allocated stalls within two weeks or face a month's suspension, and new street hawker licences ceased to be issued in 1978. Street hawker numbers fell to around 4,000 within five years and to roughly 1,000 by 1983, and the last street hawker was relocated in 1986 (National Archives of Singapore 2026).

A2 Representativeness

To assess the representativeness of our sample, we compare characteristics of survey respondents to vendors who were recruited but did not complete the survey. Table A2 presents this comparison. Survey respondents differ from non-respondents on two observable characteristics. Participants are more likely to sell produce (59.2% vs 49.5%), and less likely to sell cooked food and drinks. Participants are also slightly less likely to be male (88.3% vs 91%). Age and the possession of stalls or carts are balanced across the two groups. These patterns are consistent with two features of the data collection process. First, street food vendors require more continuous attention to preparation and service than produce vendors, leaving them with less downtime to participate in a 45-minute survey, plausibly explaining their lower participation rate. Second, in husband-and-wife operations, recruited husbands frequently delegated the main survey to their wives, which likely accounts for the lower share of male respondents among participants relative to recruits. Taken together, the differences in sample composition reflect operational and household constraints on survey participation rather than systematic selection on the economic or locational characteristics that are central to this paper's analysis.

A3 Additional Tables and Figures

A4 Balance for the Information Experiment

Table A2: Balance Table for Participants and Non-participants

Variable	(1) Non-Participants	(2) Participants	(3) Difference
Age	37.687	38.636	0.949
	(12.298)	(12.756)	(0.584)
Male	0.910	0.883	-0.027*
	(0.287)	(0.322)	(0.015)
Fresh Produce	0.495	0.592	0.097***
	(0.500)	(0.492)	(0.023)
Cooked Food/Drinks	0.505	0.408	-0.097***
	(0.500)	(0.492)	(0.023)
On Main Roads or streets	0.887	0.899	0.012
	(0.317)	(0.302)	(0.014)
On Pedestrian Walkways	0.113	0.101	-0.012
	(0.317)	(0.302)	(0.014)
Have Stalls/Carts	0.856	0.851	-0.005
	(0.351)	(0.356)	(0.016)
On the Floor	0.142	0.149	0.007
	(0.349)	(0.356)	(0.016)
Observations	564	2,922	3,486

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

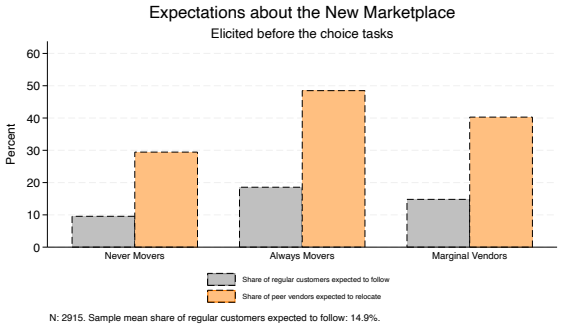
Table A3: Differences between Mover Types

	Yesterday's Profit	Weekly Profit	Legal Rights	Cognitive Ability
Never-movers	-32.52	-214.9	-0.678***	0.00464
	(24.40)	(176.3)	(0.0567)	(0.0380)
Marginal vendors	78.76***	697.0***	-0.175***	0.0819**
	(28.60)	(203.6)	(0.0483)	(0.0323)
Always-mover Mean	610.68	4126.80	2.90	2.09
Observations	2,922	2,922	2,922	2,922

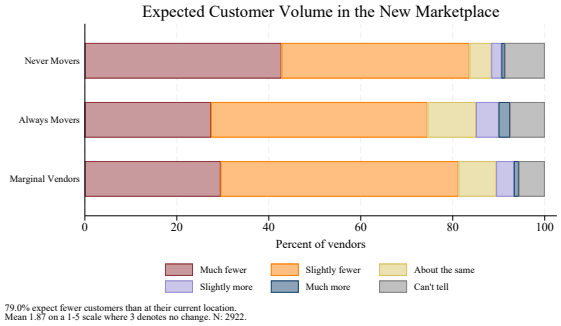
Note: Legal rights importance measured on a 1–5 scale. Cognitive ability is a sum of scores to three questions: math, pattern recognition and word recall. Monetary values in INR.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure A2: Expected Beliefs About the New Marketplace

(a) Share of regular customers and vendors



(b) Customer volume



Note: Beliefs are elicited prior to the choice tasks. Sample size for (a) is 2,915 instead of 2,922 because 7 participants refused to answer those survey questions.

Table A4: Balance Table For Information Experiment

Variable	(1) Control Group	(2) Treatment Group	(3) Difference
Age	38.39	38.88	0.49
	(12.63)	(12.87)	(0.47)
Education	6.08	6.08	-0.00
	(4.77)	(4.75)	(0.18)
Male	0.88	0.88	0.00
	(0.32)	(0.32)	(0.01)
Married	0.82	0.82	-0.00
	(0.38)	(0.38)	(0.01)
Single	0.16	0.16	0.00
	(0.36)	(0.36)	(0.01)
Widowed/Divorced	0.02	0.02	0.00
	(0.14)	(0.15)	(0.01)
Backward Caste	0.88	0.86	-0.02
	(0.33)	(0.35)	(0.01)
Scheduled Caste/Tribe	0.09	0.10	0.01
	(0.29)	(0.31)	(0.01)
General	0.03	0.04	0.00
	(0.18)	(0.20)	(0.01)
Fresh Produce	0.60	0.59	-0.01
	(0.49)	(0.49)	(0.02)
Cooked Food/Drinks	0.40	0.41	0.01
	(0.49)	(0.49)	(0.02)
On Main Roads or streets	0.90	0.90	-0.00
	(0.30)	(0.30)	(0.01)
On Pedestrian Walkways	0.10	0.10	0.00
	(0.30)	(0.30)	(0.01)
Have Stalls/Carts	0.85	0.86	0.01
	(0.36)	(0.35)	(0.01)
On the Floor	0.15	0.14	-0.01
	(0.36)	(0.35)	(0.01)
Work 7 days a week	0.95	0.94	-0.00
	(0.22)	(0.23)	(0.01)
Hours of Operation	9.19	9.25	0.06
	(3.63)	(3.70)	(0.14)
Have employees	0.30	0.33	0.02
	(0.46)	(0.47)	(0.02)
Yesterday's Sales (INR)	2,927.72	3,039.17	111.45
	(2,477.57)	(3,211.32)	(106.40)
Yesterday's Profits (INR)	612.02	624.78	12.75
	(586.92)	(662.10)	(23.18)
Weekly Sales (INR)	18,493.02	20,014.04	1,521.03*
	(15,425.35)	(28,443.03)	(851.26)
Weekly Profits (INR)	4,270.38	4,249.35	-21.03
	(3,992.45)	(4,118.06)	(150.13)
Own Other Businesses	0.15	0.14	-0.02
	(0.36)	(0.34)	(0.01)
Never Changed Marketplaces	0.99	0.99	-0.00
	(0.08)	(0.08)	(0.00)

Upto 1 year	0.06	0.07	0.01
	(0.23)	(0.25)	(0.01)
1 to 5 years	0.21	0.21	-0.01
	(0.41)	(0.40)	(0.02)
5 to 10 years	0.21	0.19	-0.02
	(0.41)	(0.39)	(0.01)
More than 10 years	0.52	0.53	0.01
	(0.50)	(0.50)	(0.02)
Policy Awareness	0.08	0.07	-0.01
	(0.27)	(0.26)	(0.01)
Risk Aversion	3.57	3.62	0.05
	(1.72)	(1.73)	(0.06)
Observations	1,432	1,490	2,922

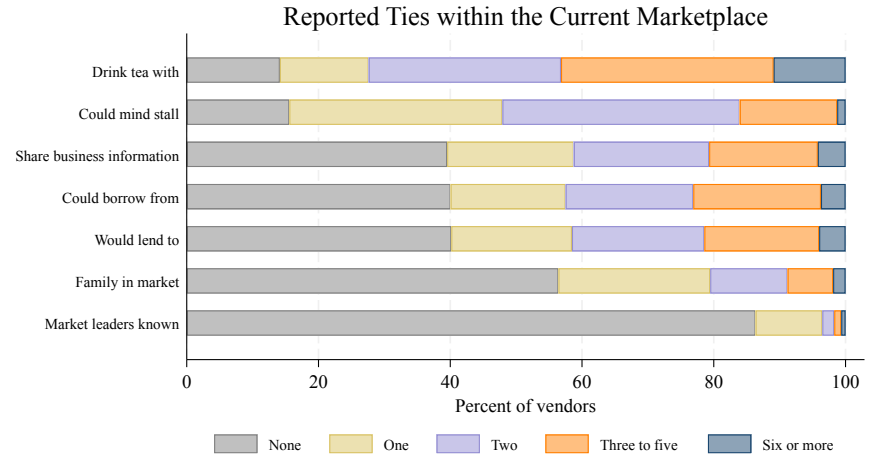
Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A5 Within-Marketplace Ties

Vendors were asked how many others in their marketplace they drink tea with, could ask to mind their stall, share business information with, could borrow from, would lend to, and are related to, alongside whether they know of anyone occupying an informal leadership role in the market. Figure A3 reports the distribution. Social contact is widespread, with 86% naming at least one vendor they drink tea with and 84% at least one who could mind their stall. Economic relations are less common and are reported in similar numbers in each direction, with 60% naming at least one vendor they could borrow from and 60% at least one they would lend to. The counts are small relative to a median enumerated market size of 50 vendors.

These record how many vendors a respondent holds each relation with and not which ones, so no network statistic beyond degree is recoverable. They are bounded by the current marketplace, so ties elsewhere in the city and relations with customers or suppliers are also not recorded.

Figure A3: Reported Social Ties Within Marketplace

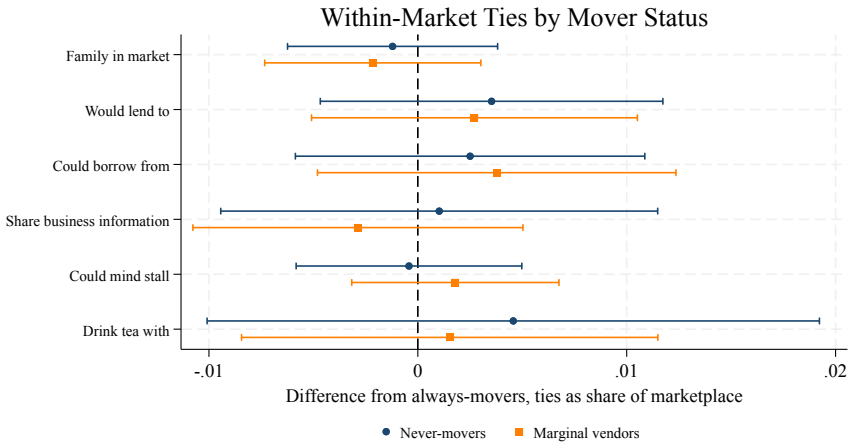


Counts of other vendors in the same marketplace with whom the respondent holds each relation.
N: 2922. Median enumerated market size is 50 vendors.

Figure A4 reports differences by mover type. Each estimate is from a separate regression of one tie share on mover type, with product category and marketplace fixed effects and standard errors clustered at the marketplace level, and always-movers as the omitted category. All twelve

intervals contain zero, with point estimates under half a percentage point of the marketplace, though the intervals do not exclude differences of up to about two percentage points.

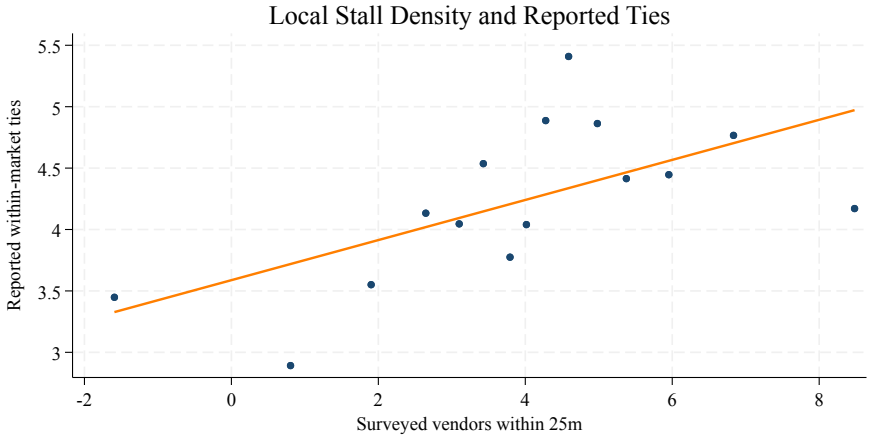
Figure A4: Differences in Reported Ties by Mover Status



Coefficients from separate OLS regressions of each tie share on mover type, with product category and marketplace fixed effects and standard errors clustered at the marketplace level.
Bars are 95% confidence intervals. Always-movers are the omitted category. N: 2922.

Figure A5 relates the counts to local stall density. Within marketplaces, a vendor with one additional surveyed neighbour within 25 metres reports 0.162 more ties (se 0.069, $p = 0.020$, N = 2,712), where ties are the largest count across the tea, information, borrowing, lending and stall-minding relations and are therefore a lower bound on the number of distinct vendors known. The density counts the vendors surveyed and not the actual number of the vendors within a radius of 25 meters, so the coefficient is per surveyed neighbour. The association is reported as a consistency check on the counts rather than as an estimate of how ties are formed.

Figure A5: Local Stall Density and Reported Ties



Binned means over 15 equal-sized bins, with marketplace fixed effects partialled out of both axes.
Slope 0.162 (se 0.069, p 0.020), N = 2712.
Ties are the largest count across the tea, information, borrowing, lending and stall-minding relations, a lower bound on distinct vendors known. Density counts surveyed vendors only.

A6 Experimental Design Details

A6.1 Scenario Construction

The choice scenarios for this stated preferences experiment were constructed through a systematic filtering process applied to a full factorial design. We began by generating all possible combinations of six attributes, each with five levels, creating a complete factorial design of 15,625 scenarios. To ensure realistic and cognitively manageable choice tasks, we applied two key filtering criteria. First, we identified and removed all weakly dominating scenarios—alternatives that were unambiguously better than the status quo across all attributes with at least one attribute strictly better—as such scenarios provide limited information about vendor trade-offs. We retained only "mixed" scenarios where some attributes were improvements while others represented trade-offs, along with the status quo scenario and scenarios where only rent varied from current conditions. Second, to maintain respondent comprehension and reduce cognitive burden, we restricted scenarios to those where a maximum of three attributes differed from the status quo, ensuring participants would face manageable comparisons rather than overwhelming changes across all six dimensions simultaneously.

From the pool of valid scenarios that satisfied both filtering criteria, we randomly selected 150 unique choice scenarios using a fixed random seed to ensure reproducibility. These 150 scenarios were then systematically organised into 25 blocks, with each block containing six choice tasks. Each respondent was assigned to one block, meaning they faced six binary choices comparing a new market alternative (described by varying attribute levels) against their current market (the status quo). This blocking strategy balanced the need for sufficient statistical power across the full sample with the practical constraint of minimising respondent fatigue. The random selection and assignment process ensured that the chosen scenarios maintained variation in attribute levels while avoiding systematic biases.

Table A5 reports pairwise correlations between attribute levels in the design matrix. All correlations are below 0.2, confirming that the design supports independent estimation of attribute effects without multicollinearity concerns.

Table A5: Correlation Matrix of Experimental Design Attributes

	Market Rent	Number of Spots	Commuting Distance	Relocating Vendors	Competitors	Customers
Market Rent	1.00					
Number of Spots	-0.12	1.00				
Commuting Distance	-0.19	-0.08	1.00			
Relocating Vendors	0.08	-0.11	-0.15	1.00		
Competitors	0.11	-0.02	-0.11	-0.12	1.00	
Customers	-0.04	0.04	0.19	-0.12	0.05	1.00

Note: Correlations calculated using coded attribute levels (1-5) from the experimental design matrix.

A6.2 Bias Mitigation

Hypothetical bias is mitigated by two design choices: personalising the status quo to each vendor's verified current conditions (Taylor and Cummings 1999), and informing vendors at the outset that their responses would directly inform actual PMC policy decisions, increasing incentive compatibility.

Inattention bias is addressed by including a rent attention check in the instrument. Approximately 4.2% of respondents failed this check, suggesting inattention is not a problem in the

sample.

Survey fatigue is addressed by (a) the ordering of the information treatment, which ensures the DiD specification's time trend μ absorbs any drift in response patterns from fatigue, and (b) a risk aversion task implemented as an incentivised game between choice modules to maintain engagement.

A6.3 Sample Stated Preferences Vignette

Figure A6 reproduces a sample choice task as presented to respondents. Given the low literacy levels among vendors, each attribute was represented using visual aids alongside verbal descriptions read aloud by the enumerator. The status quo column reflected each vendor's self-reported current conditions, elicited in the survey module immediately preceding the choice tasks.

Enumerator Script for Choice Tasks

The following script was read aloud by enumerators before each choice task was presented. The script was standardised across all enumerators and all respondents.

> *For the rest of this exercise, we want you to imagine that you are being offered a stall in a newly established marketplace and you need to choose whether you want to stay in the current marketplace or you want to move to this new market …*
>
> *…The new market offered does not differ from your current market except for the characteristics listed here. Imagine that the amount of the products that you can store, and transportation costs for products remain the same in the new marketplace …*
>
> *…In this exercise, we want you to imagine that each card represents a new scenario of the world. The only options available to you are the ones shown on the card. We want you to tell us which option you prefer out of the two. There are no "right" or "wrong" answers- you should only choose the option that you like better (even if you may not like any of them).*

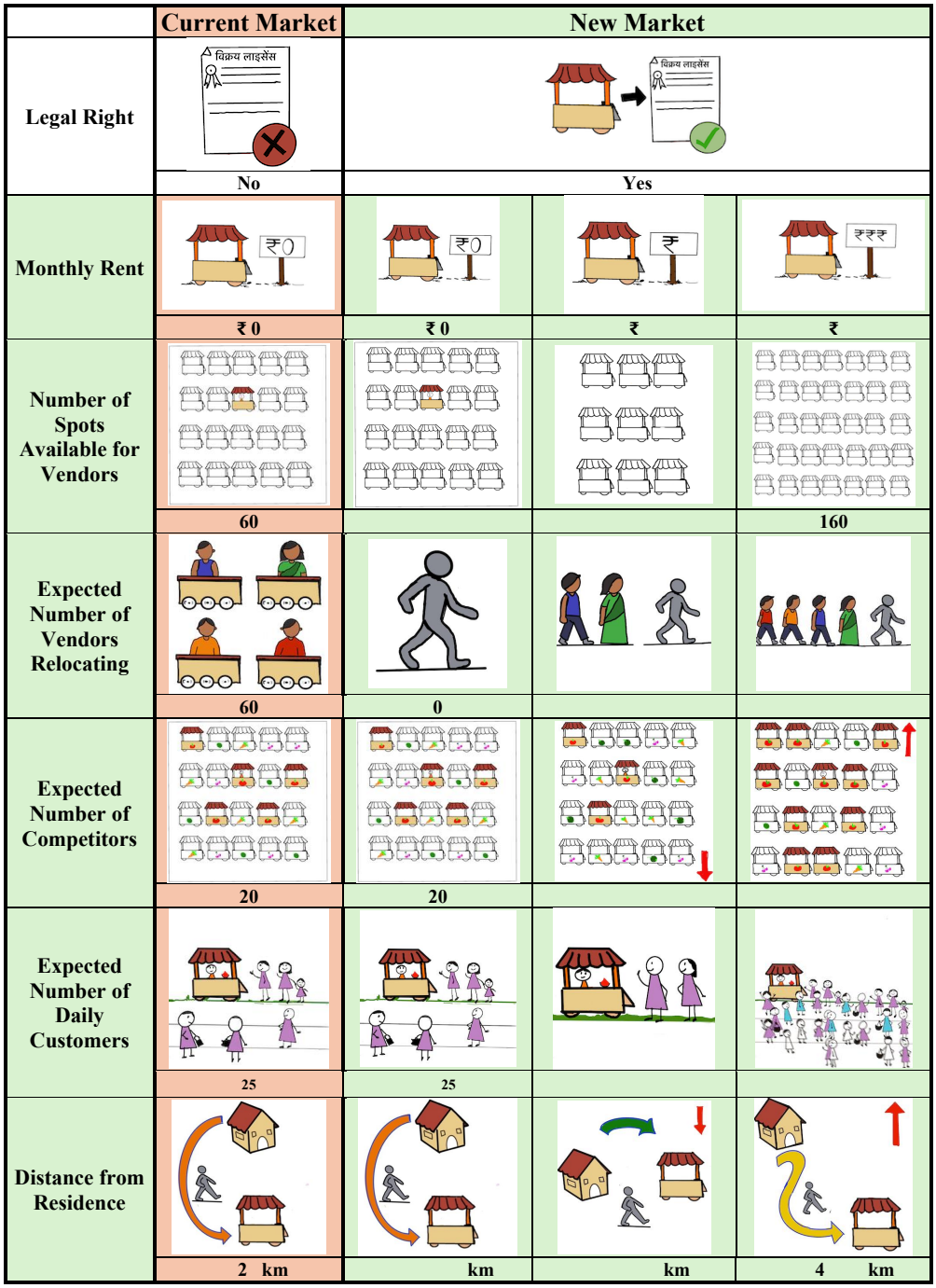
The six attributes presented in each choice task, and their possible values, are described in Table 2.

A6.4 Information Treatment Text

Control group respondents received no information and proceeded directly to round 4. The following was read aloud by enumerators to treatment group vendors immediately after round 3:

> *Let me tell you about a policy that concerns businesses like yours. It is called the Street Vendors Act of 2014. Under this act businesses like yours are offered spaces in specified vending areas with formal stalls to sell products. This is a national level policy and after its complete implementation, street vendors will not be allowed to work in unauthorized vending areas and will face fines if found to do so. For example, in Chandigarh, the municipality has situated several vendors in new marketplaces. News in December 2024 reported that about 3,000 vendors who didn't abide by the policy are not being allowed to vend in their old location in Chandigarh. The vendors who choose to move to the new marketplace will not have the option to return to their old location once the policy has been fully implemented.*

Figure A6: Sample Stated Preferences Choice Task



Notes: The choice task presents two alternatives: the vendor's current informal location (left column) and a hypothetical formal vending zone (right column with 3 sub-columns). Attribute values in the vending zone column vary across choice tasks according to the experimental design described in Section 3.1. The three sub-columns are used to depict three scenario possibilities: values remain at status-quo levels, decrease or increase. Visual aids replace text-based comparisons to accommodate low-literacy respondents. The practice round, which precedes the five analytical rounds, used a similar format and is excluded from all estimation.

A7 Robustness

This section reports the checks behind the specification, the estimation sample, and the inference. They are ordered by what the estimates depend on most: the functional form comes first, because the paper's claim is that costs bind at thresholds rather than at the margin, so the form is itself the finding; the sample second, because it is defined on the outcome and therefore needs defending; inference and the estimator last.

Functional form. Table A6 sets out how the adopted specification was arrived at, comparing it against the alternatives we considered on a common metric. Entering each attribute as free level dummies, in Table A7, rejects linearity for rent, distance and capacity. The rent levels are the reason for the threshold: relative to a zone charging no rent, a rent of one per cent of monthly sales carries a coefficient of 0.099 (s.e. 0.144), indistinguishable from zero, while two per cent carries -1.140 and eight per cent -2.618. Rent does not begin to bite until it is large relative to turnover. The distance dummies are the basis for the one-off term: any change in the commute is penalised, at -0.906 for one kilometre further and -0.291 (s.e. 0.214) for one kilometre closer, and the increments beyond the first kilometre are smaller than the first.

Six functional forms for rent were estimated on the same 2,337 choice occasions, reported in Table A8. A threshold at one per cent of sales plus a linear term in the *excess above it* fits best ($\log L = -1344.96$), ahead of a linear share (-1354.70) and a threshold at any positive rent (-1354.28). The knot was searched over the design levels and one per cent won each time. An absolute threshold at INR 450, profiled from the data, fits 22 log-likelihood points worse than the proportional one (Table A6), so vendors appear to ignore a rent that is small relative to their turnover rather than one that is small in absolute rupees.

One claim requires care. Because the knot was searched, it costs a degree of freedom. Charged for, the adopted form and a completely unrestricted set of rent level dummies are *indistinguishable* on AIC (2729.91 against 2729.58); the adopted form wins on BIC by 5.43. The defensible statement is that imposing the threshold gives up 1.16 log-likelihood points against letting all four rent levels go free, not that it dominates on every criterion.

A second restriction is rejected outright. The adopted form assumes rent enters exactly as a share of sales. Interacting the slope with log sales (Table A9) gives an implied exponent of $\hat{\gamma} = 0.574$ (s.e. 0.152, CI $[0.276, 0.872]$), excluding both the share convention ($\gamma = 1$) and the rupee convention ($\gamma = 0$). Willingness to pay therefore rises with sales, but less steeply than proportionality implies: the p10–p90 spread is 3.9-fold rather than 7.1-fold, while the median is unaffected. We retain the share form for interpretability and report the median from it, taking the spread across the sales distribution from the unrestricted model.

Table A6: Specification History: Headline Quantities on a Common Metric

Specification		$\log L$	Max. acceptable rent (INR'000) p10 sales	p50 sales	p90 sales	At median sales Move (INR'000)	Per km (INR'000)	Break-even (km)
M0	linear	-1398.37	2.92	2.92	2.92	—	-1.32	2.21
M1	+ fixed-cost distance	-1388.86	3.24	3.24	3.24	-1.82	-0.91	1.56
M2	+ share rent	-1368.50	0.79	2.10	5.61	—	-0.83	2.53
M3	both	-1354.70	0.87	2.33	6.22	-1.28	-0.54	1.94
M4	1% threshold, rupee slope	-1345.27	1.27	1.68	2.76	-3.32	-1.60	2.00
M5	**1% threshold, share slope**	-1344.96	0.71	1.91	5.10	-1.86	-0.86	1.99
M5a	absolute threshold	-1367.24	2.72	2.72	2.72	-3.66	-1.56	2.00

Note: All seven rows use the same 2,337 choice occasions. The maximum acceptable rent is what a vendor would pay for a zone matching their current pitch on every attribute except rent: a constant in M0, M1 and M5a, proportional to sales in M2, M3 and M5, and affine in sales in M4. Where the specification measures rent as a share, the move cost and the per-kilometre rate are converted at median monthly sales among marginal vendors, INR 65,000; the p10 and p90 columns are evaluated at the tenth and ninetieth percentiles of the same distribution, INR 24,267 and INR 173,333. The break-even change in commute needs no conversion, since rent cancels out of it, and is directly comparable across every row. M0 and M2 have no fixed cost of moving. Log-likelihoods are unclustered, so they are genuine rather than quasi-likelihoods.

Table A7: Attribute Levels as Dummies: Distance, Rent and Capacity

	(1)		(2)		(3)		(4)		(5)	
Monthly Rent (INR'000)	-0.324	***	-0.337	***	-0.338	***			-0.318	***
	(0.040)		(0.042)		(0.042)				(0.039)	
Vendor Capacity (10s)	-0.011		-0.012		-0.012		-0.006			
	(0.007)		(0.008)		(0.008)		(0.007)			
Relocating Vendors (%)	0.002	**	0.002	*	0.002	*	0.003	***	0.002	**
	(0.001)		(0.001)		(0.001)		(0.001)		(0.001)	
Competitors (10s)	-0.051		-0.048		-0.047		-0.057		-0.051	
	(0.060)		(0.060)		(0.061)		(0.066)		(0.061)	
Daily Customers (10s)	0.088	***	0.086	***	0.085	***	0.094	**	0.103	***
	(0.034)		(0.032)		(0.033)		(0.038)		(0.035)	
Distance from Residence (kms)	-0.429	***					-0.463	***	-0.417	***
	(0.037)						(0.037)		(0.037)	
dist = -1			-0.291		-0.341					
			(0.214)		(0.230)					
dist = 1			-0.906	***	-0.907	***				
			(0.163)		(0.163)					
dist = 3			-1.566	***	-1.573	***				
			(0.143)		(0.142)					
dist = 5			-2.134	***	-2.141	***				
			(0.196)		(0.194)					
rent = 1							0.099			
							(0.144)			
rent = 2							-1.140	***		
							(0.157)			
rent = 4							-1.470	***		
							(0.156)			
rent = 8							-2.618	***		
							(0.246)			
spots = 20									0.267	**
									(0.131)	
spots = 40									-0.271	*
									(0.153)	
spots = 80									0.132	
									(0.125)	
spots = 160									-0.423	**
									(0.179)	
Product Category										
Vegetables	0.179	**	0.203	**	0.193	**	-0.001		0.179	**
	(0.089)		(0.088)		(0.087)		(0.093)		(0.088)	
Fruits	0.139	*	0.153	**	0.148	**	-0.060		0.138	*
	(0.075)		(0.076)		(0.075)		(0.080)		(0.079)	
Status-quo Commuting Distance					0.025					
					(0.022)					
Status-quo Market Capacity									0.005	
									(0.005)	
Intercept	0.656		0.779	*	0.754	*	0.680		0.510	
	(0.411)		(0.421)		(0.420)		(0.438)		(0.421)	
Number of observations	4674		4674		4674		4674		4674	

Note: *** p<.01, ** p<.05, * p<.1. Column (1) reproduces the preferred linear specification. Columns (2) to (5) replace one linear attribute term with level indicators, omitting the no-change level. Distance levels are the realised change in commuting distance: negative changes that would have fallen below 1 km were floored at 1 km by design, so the realised change rather than the assigned one is the attribute the vendor faced. Column (3) adds the vendor's own baseline distance as a case variable, since the reference category pools assigned no-change with floored cases. Where the assigned level is not the realised deviation, the vendor's own level is retained as a case variable. Standard errors are clustered at the marketplace level.

Table A8: Estimations at Various Rent Thresholds

			Knot taken as given		Knot charged	
Rent form	$\log L$	k	AIC	BIC	AIC	BIC
Knot at 0% (any rent) + linear	-1354.28	19	2746.56	2855.94	2748.56	2863.69
Knot at 1% + linear above (M5)	-1344.96	19	2727.91	2837.29	2729.91	2845.04
Knot at 2% + linear above	-1384.41	19	2806.82	2916.19	2808.82	2923.95
Knot at 4% (degenerate)	-1442.85	18	2921.69	3025.31	2923.69	3033.07
Free level dummies	-1343.79	21	2729.58	2850.47	2729.58	2850.47
Linear in share (M3)	-1354.70	18	2745.40	2849.02	2745.40	2849.02

The estimation sample. Estimates are reported for marginal vendors. Because that group is defined by how a vendor answered, the restriction requires justification rather than assertion. Two things support it. The marginal vendor sample is exactly what a vendor fixed-effects conditional logit retains, since that estimator discards respondents whose choices do not vary and contributes nothing to the likelihood, so the restriction is a property of the estimator rather than a choice made after seeing the results. And the three mover types differ on characteristics fixed before the experiment — valuation of legal rights ($p < 0.0001$), profit ($p = 0.055$) and tenure at the current pitch ($p = 0.017$), from a joint test across the three types on the regressions in Table A3 — so they are distinct populations rather than draws from one.

The consequence for the estimates is large and one-directional (Table A10). The one-off cost of moving is -0.714 among marginal vendors and -0.088 on the full pre-treatment sample, and the joint test that the two distance terms are common across the two groups gives $\chi^2(2) = 73.86$. That moves the break-even change in commute from 1.99 to 3.89 kilometres. Pooling attenuates a coefficient identified only off vendors whose choices respond; it is dilution rather than a competing estimate, and the counterfactual is reported at the marginal value.

Allowing a random zone intercept on the full sample separates the sample from the estimator (Table A11). The per-kilometre rate generalises — -1.326 on marginal vendors, -1.208 on the full sample with the same estimator, -1.237 with random intercepts. The one-off cost does not: -2.869 becomes -0.576 and recovers only to -0.964 once intercept heterogeneity is allowed. Had the excluded groups differed only in levels, a random intercept would have absorbed them. It does not, so they plausibly differ in slopes as well, and we do not claim the excluded vendors are simply intercept outliers.

Table A9: Does the Share Slope Vary with Sales?

	(1)		(2)		(3)		(4)	
$\mathbb{1}$[Rent > 1%]	-0.889	***	-0.890	***	-0.878	***	-0.884	***
	(0.149)		(0.147)		(0.147)		(0.147)	
Rent above 1% (pp of sales)	-0.249	***	-0.249	***	-0.252	***	-0.251	***
	(0.042)		(0.041)		(0.040)		(0.040)	
Vendor Capacity (10s)	-0.007		-0.008		-0.008		-0.008	
	(0.007)		(0.007)		(0.007)		(0.007)	
Relocating Vendors (%)	0.002	**	0.002	**	0.002	**	0.002	**
	(0.001)		(0.001)		(0.001)		(0.001)	
Competitors (10s)	-0.053		-0.051		-0.051		-0.051	
	(0.066)		(0.065)		(0.064)		(0.064)	
Daily Customers (10s)	0.099	***	0.098	***	0.098	***	0.098	***
	(0.035)		(0.035)		(0.035)		(0.035)	
$\mathbb{1}[\Delta$Commuting Distance $\neq 0]$	-0.714	***	-0.709	***	-0.714	***	-0.712	***
	(0.149)		(0.149)		(0.149)		(0.149)	
Distance from Residence (kms)	-0.330	***	-0.332	***	-0.332	***	-0.332	***
	(0.043)		(0.043)		(0.043)		(0.043)	
Rent above 1% x log sales			-0.106	***			-0.051	
			(0.032)				(0.051)	
Rent-above-1% indicator x log sales					-0.325	***	-0.198	
					(0.108)		(0.179)	
Product Category								
Vegetables	0.015		0.079		0.095		0.095	
	(0.092)		(0.091)		(0.093)		(0.093)	
Fruits	-0.054		0.003		0.021		0.019	
	(0.081)		(0.083)		(0.086)		(0.086)	
Legal Right Importance								
A little important	0.263		0.223		0.176		0.190	
	(0.440)		(0.419)		(0.438)		(0.430)	
Somewhat important	0.115		0.105		0.062		0.077	
	(0.427)		(0.405)		(0.422)		(0.415)	
Very important	0.460		0.454		0.413		0.428	
	(0.429)		(0.407)		(0.423)		(0.416)	
Extremely important	0.533		0.548		0.507		0.524	
	(0.453)		(0.430)		(0.446)		(0.439)	
Years in Business								
1 to 5 years	0.187		0.204		0.210		0.209	
	(0.149)		(0.148)		(0.150)		(0.149)	
5 to 10 years	0.073		0.092		0.094		0.095	
	(0.156)		(0.158)		(0.159)		(0.159)	
More than 10 years	0.115		0.140		0.149		0.148	
	(0.142)		(0.140)		(0.142)		(0.142)	
Risk Aversion	-0.004		-0.004		-0.003		-0.003	
	(0.019)		(0.019)		(0.019)		(0.019)	
Intercept	0.874	**	0.815	**	0.841	**	0.826	**
	(0.436)		(0.409)		(0.423)		(0.415)	
Number of observations	4674		4674		4674		4674	

Note: *** p<.01, ** p<.05, * p<.1. Column (1) is M5. Column (2) adds the rent slope interacted with log monthly sales, centred at the median; its coefficient is zero if and only if rent enters exactly as a share of sales. Column (3) instead interacts the threshold indicator, which is already written as one per cent of the vendor's own sales and so should carry no residual gradient. Column (4) includes both. Standard errors are clustered at the marketplace level.

Table A10: Is the One-Off Cost of Moving Stable?

	(1)		(2)		(3)		(4)	
$\mathbb{1}$[Rent > 1%]	-0.889	***	-0.366	***	-0.338	***	-0.797	***
	(0.149)		(0.071)		(0.072)		(0.156)	
Rent above 1% (pp of sales)	-0.249	***	-0.152	***	-0.160	***	-0.265	***
	(0.042)		(0.024)		(0.024)		(0.044)	
Vendor Capacity (10s)	-0.007		0.021	***	0.021	***	-0.005	
	(0.007)		(0.007)		(0.006)		(0.007)	
Relocating Vendors (%)	0.002	**	0.010	***	0.010	***	0.003	***
	(0.001)		(0.001)		(0.001)		(0.001)	
Competitors (10s)	-0.053		-0.074	**	-0.072	**	-0.057	
	(0.066)		(0.031)		(0.030)		(0.060)	
Daily Customers (10s)	0.099	***	0.052	***	0.058	***	0.103	***
	(0.035)		(0.019)		(0.019)		(0.036)	
$\mathbb{1}[\Delta\text{Commuting Distance} \neq 0]$	-0.714	***	-0.088		0.040		-0.623	***
	(0.149)		(0.074)		(0.079)		(0.196)	
Distance from Residence (kms)	-0.330	***	-0.184	***	-0.113	***	-0.295	***
	(0.043)		(0.025)		(0.026)		(0.046)	
Fixed cost x marginal vendor					-0.506	***		
					(0.162)			
Per km x marginal vendor					-0.223	***		
					(0.046)			
Fixed cost x task 3							-0.624	***
							(0.171)	
Fixed cost x task 4							0.401	*
							(0.205)	
$\mathbb{1}[i \in$ Marginal Vendors]					0.155			
					(0.100)			
Intercept	0.874	**	-2.767	***	-2.861	***	0.735	*
	(0.436)		(0.405)		(0.407)		(0.413)	
Number of observations	4674		17532		17532		4674	

Note: *** p<.01, ** p<.05, * p<.1. Column (1) is the adopted specification on marginal vendors, the sample the paper reports. Column (2) is the same specification on all pre-treatment cases. Column (3) pools them and interacts both distance terms with an indicator for being a marginal vendor, which is the test of whether the difference between the first two columns is real; the samples are nested, so reading their confidence intervals against one another would not be. Column (4) interacts the one-off cost with the choice task, testing whether it drifts across the three pre-treatment rounds. Standard errors are clustered at the marketplace level throughout. Product category, legal right importance, years in business and risk aversion are included as controls in all 4 columns but not shown.

Table A11: Random Zone Intercept on the Full Sample

	(1)		(2)		(3)		(4)	
$\mathbb{1}$[Rent > 1%]	-0.889	***	-0.366	***	-0.864	***	-0.864	***
	(0.149)		(0.071)		(0.148)		(0.148)	
Rent above 1% (pp of sales)	-0.249	***	-0.152	***	-0.390	***	-0.390	***
	(0.042)		(0.024)		(0.049)		(0.049)	
Vendor Capacity (10s)	-0.007		0.021	***	0.021	**	0.021	**
	(0.007)		(0.007)		(0.010)		(0.010)	
Relocating Vendors (%)	0.002	**	0.010	***	0.015	***	0.015	***
	(0.001)		(0.001)		(0.002)		(0.002)	
Competitors (10s)	-0.053		-0.074	**	-0.159	***	-0.159	***
	(0.066)		(0.031)		(0.047)		(0.047)	
Daily Customers (10s)	0.099	***	0.052	***	0.144	***	0.144	***
	(0.035)		(0.019)		(0.038)		(0.038)	
$\mathbb{1}[\Delta$Commuting Distance $\neq 0]$	-0.714	***	-0.088		-0.376	**	-0.376	**
	(0.149)		(0.074)		(0.153)		(0.153)	
Distance from Residence (kms)	-0.330	***	-0.184	***	-0.483	***	-0.483	***
	(0.043)		(0.025)		(0.055)		(0.055)	
Vending zone alternative					-7.020	***	-7.021	***
					(0.970)		(0.973)	
Product Category								
Vegetables	0.015		-0.015		-0.060		-0.060	
	(0.092)		(0.087)		(0.245)		(0.245)	
Fruits	-0.054		0.150	*	0.393		0.393	
	(0.081)		(0.091)		(0.258)		(0.258)	
Legal Right Importance								
A little important	0.263		2.294	***	6.148	***	6.149	***
	(0.440)		(0.392)		(0.945)		(0.946)	
Somewhat important	0.115		2.267	***	6.076	***	6.077	***
	(0.427)		(0.385)		(0.900)		(0.901)	
Very important	0.460		3.167	***	8.617	***	8.618	***
	(0.429)		(0.385)		(0.902)		(0.903)	
Extremely important	0.533		3.963	***	10.839	***	10.840	***
	(0.453)		(0.401)		(1.000)		(1.001)	
Years in Business								
1 to 5 years	0.187		0.158		0.398		0.399	
	(0.149)		(0.148)		(0.415)		(0.415)	
5 to 10 years	0.073		0.115		0.311		0.312	
	(0.156)		(0.166)		(0.459)		(0.460)	
More than 10 years	0.115		0.256	*	0.688		0.688	
	(0.142)		(0.150)		(0.425)		(0.425)	
Risk Aversion	-0.004		0.008		0.018		0.018	
	(0.019)		(0.019)		(0.053)		(0.053)	
Intercept	0.874	**	-2.767	***				
	(0.436)		(0.405)					
sd(zone)					4.277		4.277	
					(0.179)		(0.179)	
Number of observations	4674		17532		17532		17532	

Note: *** p<.01, ** p<.05, * p<.1. Every column is the adopted specification M5: rent is ignored at or below one per cent of monthly sales and linear in the excess above it, and distance enters as a one-off cost of any change plus a linear per-kilometre term. Column (1) is the conditional logit on marginal vendors, the specification the paper reports. Column (2) is the same estimator on the full pre-treatment sample, included so that the sample and the estimator can be separated. Columns (3) and (4) are panel mixed logit on the full pre-treatment sample with a random coefficient on the vending zone indicator, estimated by simulated maximum likelihood with Hammersley sequences at 200 and 400 integration points. The alternative-specific constant is suppressed because its role is taken by the mean of the random zone coefficient. Coefficients are not directly comparable across columns because the scale parameter differs; compare the willingness-to-pay ratios, which are scale free. The zone intercept is not comparable either: the full sample retains never-movers, so the average propensity to accept is far more negative by construction. Standard errors are clustered at the marketplace level.

Inference and design mechanics. Standard errors are compared across conventional maximum likelihood, robust, vendor-clustered and marketplace-clustered estimators, and against a pairs cluster bootstrap on marketplaces with 500 replications (Table A12). Point estimates are identical throughout and only the variance estimator differs. The reported marketplace-clustered errors sit just below the bootstrap, so inference is if anything conservative.

Balance on the realised design draws requires one distinction (Table A13). Five of twelve vendor characteristics appear unbalanced on the six assigned codes, but this detects a non-randomised regressor rather than a failure of randomisation: the relocating-vendor attribute was constructed by perturbing each vendor's own elicited belief by a randomised shock, so it correlates 0.80 with that belief. Replacing it with the randomised shock itself — the component the design actually assigned — leaves *one* of twelve characteristics significant at five per cent, risk aversion, which enters the model as a control in any case.

Two design features bound results that would otherwise be read too strongly (Tables A14 and A15). The competitor count was truncated at each zone's assigned capacity, and on the subsample where the cap does not bind competitors are significant ($p = 0.029$), so the null result on competitors reflects the design rather than indifference to rivalry. Dropping the choice occasions where an assigned reduction in distance or competitors was floored leaves the remaining coefficients unchanged. And the relocating-vendor attribute carries the vendor's own belief, so its coefficient is not identified by the experiment alone: controlling for that belief leaves the peer effect indistinguishable from zero, and the randomised shock on its own is insignificant ($p = 0.225$). We therefore treat the coordination estimate as an upper bound.

Table A12: Standard Errors under Alternative Variance Estimators

	(1)		(2)		(3)		(4)		(5)	
$\mathbb{1}$[Rent > 1%]	-0.889	***	-0.889	***	-0.889	***	-0.889	***	-0.889	***
	(0.150)		(0.149)		(0.159)		(0.149)		(0.154)	
Rent above 1% (pp of sales)	-0.249	***	-0.249	***	-0.249	***	-0.249	***	-0.249	***
	(0.042)		(0.042)		(0.042)		(0.042)		(0.044)	
Vendor Capacity (10s)	-0.007		-0.007		-0.007		-0.007		-0.007	
	(0.008)		(0.008)		(0.007)		(0.007)		(0.007)	
Relocating Vendors (%)	0.002		0.002		0.002	**	0.002	**	0.002	**
	(0.001)		(0.001)		(0.001)		(0.001)		(0.001)	
Competitors (10s)	-0.053		-0.053		-0.053		-0.053		-0.053	
	(0.044)		(0.057)		(0.065)		(0.066)		(0.086)	
Daily Customers (10s)	0.099	***	0.099	***	0.099	***	0.099	***	0.099	***
	(0.035)		(0.035)		(0.035)		(0.035)		(0.035)	
$\mathbb{1}[\Delta\text{Commuting Distance} \neq 0]$	-0.714	***	-0.714	***	-0.714	***	-0.714	***	-0.714	***
	(0.143)		(0.149)		(0.149)		(0.149)		(0.157)	
Distance from Residence (kms)	-0.330	***	-0.330	***	-0.330	***	-0.330	***	-0.330	***
	(0.038)		(0.038)		(0.039)		(0.043)		(0.045)	
Product Category										
Vegetables	0.015		0.015		0.015		0.015		0.015	
	(0.116)		(0.116)		(0.085)		(0.092)		(0.096)	
Fruits	-0.054		-0.054		-0.054		-0.054		-0.054	
	(0.116)		(0.117)		(0.085)		(0.081)		(0.081)	
Legal Right Importance										
A little important	0.263		0.263		0.263		0.263		0.263	
	(0.541)		(0.633)		(0.444)		(0.440)		(0.480)	
Somewhat important	0.115		0.115		0.115		0.115		0.115	
	(0.550)		(0.645)		(0.454)		(0.427)		(0.469)	
Very important	0.460		0.460		0.460		0.460		0.460	
	(0.529)		(0.624)		(0.438)		(0.429)		(0.472)	
Extremely important	0.533		0.533		0.533		0.533		0.533	
	(0.550)		(0.644)		(0.454)		(0.453)		(0.493)	
Years in Business										
1 to 5 years	0.187		0.187		0.187		0.187		0.187	
	(0.212)		(0.220)		(0.156)		(0.149)		(0.159)	
5 to 10 years	0.073		0.073		0.073		0.073		0.073	
	(0.216)		(0.221)		(0.154)		(0.156)		(0.166)	
More than 10 years	0.115		0.115		0.115		0.115		0.115	
	(0.200)		(0.208)		(0.149)		(0.142)		(0.152)	
Risk Aversion	-0.004		-0.004		-0.004		-0.004		-0.004	
	(0.028)		(0.028)		(0.020)		(0.019)		(0.020)	
Intercept	0.874		0.874		0.874	*	0.874	**	0.874	*
	(0.561)		(0.652)		(0.448)		(0.436)		(0.471)	
Number of observations	4674		4674		4674		4674		4674	

Note: *** p<.01, ** p<.05, * p<.1. Point estimates are identical in every column; only the variance estimator differs. Columns are conventional maximum likelihood, heteroskedasticity-robust, clustered at the vendor, clustered at the marketplace, and a pairs cluster bootstrap at the marketplace with 500 replications. Assignment was stratified within marketplace, so the marketplace is the design-relevant clustering level.

Table A13: Balance of Vendor Characteristics on Realised Design Draws

	p_OLS	p_cluster	p_mktFE	p_shock
sex	.4903838	.4031947	.3792071	.5475617
respondent_age	.0061538	.0009072	.0009596	.0689017
respondent_edu	.1429071	.0543585	.0952364	.1665147
product_category	.0264207	.0235831	.0826631	.2166159
monthly_sales	.0005856	.0006716	.0778963	.3535902
business_years	.5061056	.4151541	.4180461	.5025601
residence_proximity	.6215574	.6024006	.9970031	.7129932
footfall	.4864844	.4957738	.8448483	.6166437
policy_aware	.0790249	.2448469	.3050023	.1466384
eviction	.5697842	.3603185	.6348205	.5641294
ra_choice	.0115311	.0042251	.0082	.0063038
legal_right_importance	8.18e-08	7.15e-06	1.69e-06	.7639437

Note: Each row regresses one vendor characteristic on the vendor's mean assigned value of all six attributes across their three pre-treatment choice tasks, and reports the p-value of the joint test that all six coefficients are zero. Columns differ in the variance estimator and in the sixth regressor. The relocating-vendor attribute was built from the vendor's own elicited belief perturbed by a randomised shock, so it is not itself randomly assigned; the final column replaces it with that shock. That column is the test of the design.

Choice of estimator. Random coefficients are added one attribute at a time (Table A16). A random zone intercept buys 1,260 log-likelihood points and adding distance a further 52; everything beyond that buys nothing, and two rungs fail to converge. The ladder is identified two coefficients deep and no further.

Model choice is then settled on out-of-sample prediction rather than on a boundary-constrained variance test, since the null places the mixing variances on the boundary of the parameter space and the naive likelihood ratio test is conservative there. Holding out the control group's post-treatment rounds (Table A17), the conditional logit attains the best held-out log-likelihood (-1762.06, against -1767.59 for the random intercept), and hit rates are indistinguishable across the ladder at 0.650 to 0.652. We therefore report the conditional logit.

Unobserved heterogeneity in the zone intercept is nonetheless large. Its standard deviation is 4.28 on the logit scale with a random intercept alone and 5.47 once distance is also allowed to vary. It integrates out: at the second of those the predicted distribution of relocation probabilities has standard deviation 0.174 against 0.189 for the conditional logit, and a median of 0.599 against 0.609. Identical vendors offered an identical zone behave very differently, but the predictions here are unconditional, so we cannot say *which* vendors relocate.

Table A14: Sensitivity to the Design's Floors and Caps

	(1)		(2)		(3)		(4)	
$\mathbb{1}$[Rent > 1%]	-0.889	***	-0.909	***	-0.886	***	-0.885	***
	(0.149)		(0.156)		(0.167)		(0.150)	
Rent above 1% (pp of sales)	-0.249	***	-0.248	***	-0.263	***	-0.250	***
	(0.042)		(0.042)		(0.044)		(0.042)	
Vendor Capacity (10s)	-0.007		-0.004		-0.007		-0.004	
	(0.007)		(0.008)		(0.008)		(0.008)	
Relocating Vendors (%)	0.002	**	0.003	**	0.003	**	0.002	**
	(0.001)		(0.001)		(0.001)		(0.001)	
Competitors (10s)	-0.053		-0.302	**	-0.057		-0.021	
	(0.066)		(0.138)		(0.077)		(0.058)	
Daily Customers (10s)	0.099	***	0.103	***	0.119	***	0.098	***
	(0.035)		(0.037)		(0.039)		(0.035)	
$\mathbb{1}[\Delta$Commuting Distance $\neq 0]$	-0.714	***	-0.706	***	-0.745	***	-0.715	***
	(0.149)		(0.153)		(0.158)		(0.149)	
Distance from Residence (kms)	-0.330	***	-0.344	***	-0.336	***	-0.327	***
	(0.043)		(0.044)		(0.043)		(0.043)	
Vendor Capacity x cap binds							-0.028	
							(0.027)	
Product Category								
Vegetables	0.015		-0.085		0.053		0.010	
	(0.092)		(0.093)		(0.100)		(0.093)	
Fruits	-0.054		-0.076		0.004		-0.057	
	(0.081)		(0.082)		(0.096)		(0.082)	
Legal Right Importance								
A little important	0.263		0.187		0.308		0.253	
	(0.440)		(0.476)		(0.494)		(0.440)	
Somewhat important	0.115		0.011		0.256		0.103	
	(0.427)		(0.457)		(0.494)		(0.428)	
Very important	0.460		0.409		0.500		0.454	
	(0.429)		(0.459)		(0.494)		(0.428)	
Extremely important	0.533		0.548		0.546		0.522	
	(0.453)		(0.470)		(0.521)		(0.453)	
Years in Business								
1 to 5 years	0.187		0.165		0.109		0.189	
	(0.149)		(0.154)		(0.191)		(0.149)	
5 to 10 years	0.073		0.086		-0.027		0.076	
	(0.156)		(0.158)		(0.198)		(0.156)	
More than 10 years	0.115		0.152		0.015		0.121	
	(0.142)		(0.142)		(0.179)		(0.142)	
Risk Aversion	-0.004		-0.020		0.004		-0.005	
	(0.019)		(0.021)		(0.023)		(0.019)	
Intercept	0.874	**	0.981	**	0.870	*	0.868	**
	(0.436)		(0.443)		(0.490)		(0.437)	
Number of observations	4674		4342		3964		4674	

Note: *** p<.01, ** p<.05, * p<.1. Column (1) is the preferred specification. Column (2) drops cases where the competitor count was truncated by the zone's capacity cap, column (3) drops cases where an assigned reduction in distance or competitors was floored, and column (4) adds an interaction between capacity and an indicator for the cap binding. The cap makes competitors a deterministic function of capacity where it binds, so column (2) is the cleaner read on the capacity coefficient. Standard errors are clustered at the marketplace level.

Table A15: Identification of the Relocating-Vendor Effect

	(1)		(2)		(3)	
$\mathbb{1}$[Rent > 1%]	-0.889	***	-0.889	***	-0.889	***
	(0.149)		(0.151)		(0.148)	
Rent above 1% (pp of sales)	-0.249	***	-0.243	***	-0.249	***
	(0.042)		(0.042)		(0.041)	
Vendor Capacity (10s)	-0.007		-0.007		-0.008	
	(0.007)		(0.007)		(0.007)	
Relocating Vendors (%)	0.002	**	0.001			
	(0.001)		(0.002)			
Competitors (10s)	-0.053		-0.053		-0.052	
	(0.066)		(0.066)		(0.066)	
Daily Customers (10s)	0.099	***	0.100	***	0.102	***
	(0.035)		(0.035)		(0.035)	
$\mathbb{1}[\Delta\text{Commuting distance} \neq 0]$	-0.714	***	-0.720	***	-0.706	***
	(0.149)		(0.152)		(0.152)	
Distance from Residence (kms)	-0.330	***	-0.327	***	-0.330	***
	(0.043)		(0.043)		(0.043)	
Shock to the relocating-vendor belief (%)					0.003	
					(0.002)	
Product Category						
Vegetables	0.015		0.031		0.021	
	(0.092)		(0.089)		(0.093)	
Fruits	-0.054		-0.065		-0.052	
	(0.081)		(0.081)		(0.082)	
Legal Right Importance						
A little important	0.263		0.264		0.264	
	(0.440)		(0.446)		(0.440)	
Somewhat important	0.115		0.121		0.125	
	(0.427)		(0.432)		(0.426)	
Very important	0.460		0.467		0.484	
	(0.429)		(0.433)		(0.428)	
Extremely important	0.533		0.533		0.557	
	(0.453)		(0.458)		(0.453)	
Years in Business						
1 to 5 years	0.187		0.177		0.189	
	(0.149)		(0.148)		(0.149)	
5 to 10 years	0.073		0.079		0.086	
	(0.156)		(0.155)		(0.157)	
More than 10 years	0.115		0.110		0.127	
	(0.142)		(0.140)		(0.142)	
Risk Aversion	-0.004		-0.002		-0.005	
	(0.019)		(0.019)		(0.019)	
Expected number of other vendors relocating			0.002			
			(0.003)			
Intercept	0.874	**	0.839	*	0.936	**
	(0.436)		(0.441)		(0.437)	
Number of observations	4674		4626		4674	

Note: *** p<.01, ** p<.05, * p<.1. The relocating-vendor attribute was constructed from each vendor's own elicited belief about how many others would move, perturbed by a randomised shock of -50, -20, 0, 20 or 50 per cent. Column (1) is the preferred specification. Column (2) adds the elicited belief as a case variable, so the attribute coefficient is identified off the randomised shock alone. Column (3) replaces the attribute with the shock itself. Standard errors are clustered at the marketplace level.

Table A16: Random Coefficient Ladder

	(1)		(2)		(3)		(4)		(5)		(6)		(7)		(8)	
$\mathbb{1}\{\text{Rent}^{\%} > 1\}$	-0.366	***	-0.864	***	-1.006	***	-1.009	***	-0.811	***	-1.016	***	-0.808	***	-1.009	***
	(0.071)		(0.148)		(0.187)		(0.187)		(0.145)		(0.187)		(0.148)		(0.188)	
Rent above 1% (pp of sales)	-0.152	***	-0.390	***	-0.501	***	-0.502	***	-0.389	***	-0.500	***	-0.383	***	-0.504	***
	(0.024)		(0.049)		(0.062)		(0.062)		(0.047)		(0.062)		(0.048)		(0.063)	
Vendor Capacity (10s)	0.021	***	0.021	**	0.023	*	0.023	*	0.022	**	0.024	*	0.021	**	0.023	*
	(0.007)		(0.010)		(0.013)		(0.013)		(0.010)		(0.013)		(0.011)		(0.013)	
Relocating Vendors (%)	0.010	***	0.015	***	0.018	***	0.018	***	0.015	***	0.018	***	0.016	***	0.018	***
	(0.001)		(0.002)		(0.003)		(0.003)		(0.002)		(0.003)		(0.002)		(0.003)	
Competitors (10s)	-0.074	**	-0.159	***	-0.181	***	-0.181	***	-0.156	***	-0.217	***	-0.146	***	-0.185	***
	(0.031)		(0.047)		(0.060)		(0.058)		(0.049)		(0.072)		(0.047)		(0.060)	
Daily Customers (10s)	0.052	***	0.144	***	0.196	***	0.197	***	0.143	***	0.199	***	0.135	***	0.200	***
	(0.019)		(0.038)		(0.050)		(0.050)		(0.037)		(0.051)		(0.038)		(0.051)	
Distance from Residence (kms)	-0.184	***	-0.483	***	-0.667	***	-0.667	***	-0.496	***	-0.672	***	-0.482	***	-0.666	***
	(0.025)		(0.055)		(0.085)		(0.085)		(0.058)		(0.085)		(0.059)		(0.081)	
$\mathbb{1}[\Delta\text{Commuting distance} \neq 0]$	-0.088		-0.376	**	-0.350	*	-0.356	*	-0.299	**	-0.351	*	-0.323	**	-0.369	**
	(0.074)		(0.153)		(0.183)		(0.182)		(0.147)		(0.184)		(0.150)		(0.181)	
Vending zone alternative	-2.767	***	-7.020	***	-8.694	***	-8.621	***	-5.974	***	-8.636	***	-5.390	***	-8.784	***
	(0.405)		(0.970)		(1.263)		(1.314)		(0.814)		(1.294)		(0.769)		(1.299)	
sd(zone)			4.277		5.467		5.488		4.007		5.498		4.188		5.580	
			(0.179)		(0.306)		(0.313)		(0.157)		(0.313)		(0.167)		(0.316)	
sd(commuting distance)					0.931		0.931		0.518		0.928		0.594		0.887	
					(0.098)		(0.098)		(0.068)		(0.094)		(0.067)		(0.095)	
sd(market size)							0.029		0.004		0.030		0.007			
							(0.037)		(0.008)		(0.019)		(0.010)			
sd(competitors)									0.000		0.177		0.000			
									(0.000)		(0.051)		(0.000)			
sd(customers)											0.018		0.019			
											(0.148)		(0.012)			
sd(share vendors)													0.001			
													(0.007)			
corr(zone,commuting distance)															-0.178	
															(0.109)	
Number of observations	17532		17532		17532		17532		17532		17532		17532		17532	

Note: *** p<.01, ** p<.05, * p<.1. Every column is the adopted specification M5 – rent ignored at or below one per cent of monthly sales and linear in the excess above it, distance a one-off cost of any change plus a linear per-kilometre term – on the full pre-treatment sample. Column L0 is a conditional logit; the remaining columns are panel mixed logit with the random coefficients added one at a time, estimated by simulated maximum likelihood with Hammersley sequences at 200 integration points. The rent coefficient is fixed throughout, because a random rent coefficient would make every willingness-to-pay figure a ratio of normals with no finite moments. The fixed cost of changing the commute is also held fixed, being identified off only the 39 per cent of offers that changed the commute. Standard errors are clustered at the marketplace level.

Table A17: Model Selection on Out-of-Sample Prediction

	In sample		Held out: H1		Held out: H2		WTP	
Model	log L	df	log L	Hit	log L	Hit	Zone	Per km
M5, marginal vendors	-1344.96	19	-2000.63	0.603	-459.84	0.735	0.94	-1.33
L0 conditional logit	-5270.43	19	-1762.06	0.650	-501.56	0.654	-19.55	-1.21
L1 + random zone	-4010.34	20	-1767.59	0.652	-499.41	0.660	-19.20	-1.24
L2 + distance	-3958.31	21	-1767.20	0.652	-497.29	0.667	-18.38	-1.33
L3 + capacity	-3958.28	22	-1767.22	0.652	-497.25	0.667	-18.18	-1.33
L4 + competitors†	-3980.96	22	-1765.52	0.652	-496.52	0.662	-16.42	-1.27
L5 + customers	-3958.09	24	-1767.26	0.652	-497.16	0.667	-18.31	-1.34
L6 + relocating vendors†	-3977.97	24	-1766.30	0.650	-497.10	0.659	-15.20	-1.26
L7 zone, distance correlated	-3956.91	22	-1767.75	0.651	-497.84	0.664	-18.42	-1.32

Note: Every model is the adopted specification M5 – rent ignored at or below one per cent of monthly sales and linear in the excess above it, distance a one-off cost of any change plus a linear per-kilometre term – estimated on the pre-treatment rounds. H1 holds out all 2864 choice occasions of the control group's post-treatment rounds; H2 holds out the 810 of those belonging to marginal vendors. Controls received no information treatment, so the pre-treatment parameters should transfer. The held-out log-likelihood sums the log probability the model assigns to the alternative the vendor actually chose; the hit rate is the share of held-out occasions where that alternative was the one the model ranked first. Predictions after the mixed logit are unconditional, integrating over the mixing distribution rather than conditioning on the panel's own observed choices, so a respondent's later rounds are not used to predict themselves. The first row is estimated on marginal vendors and the rest on the full pre-treatment sample; it therefore differs in intercept calibration as well as in slopes, and H1 contains never-movers it never saw. The clean within-estimator comparison is L0 against the rungs above it. Willingness to pay is in percentage points of monthly sales; the zone column excludes case variables. † marks a rung that stopped on the iteration bound rather than converging: its coefficient vector is not a maximum, so its held-out figures are reported only to show that the breakdown is visible out of sample too.